

Homogeneous Transformations and Rigid Body Motion

Hassan Jaleel

Department of Electrical Engineering

Syed Babar Ali School of Science and Engineering

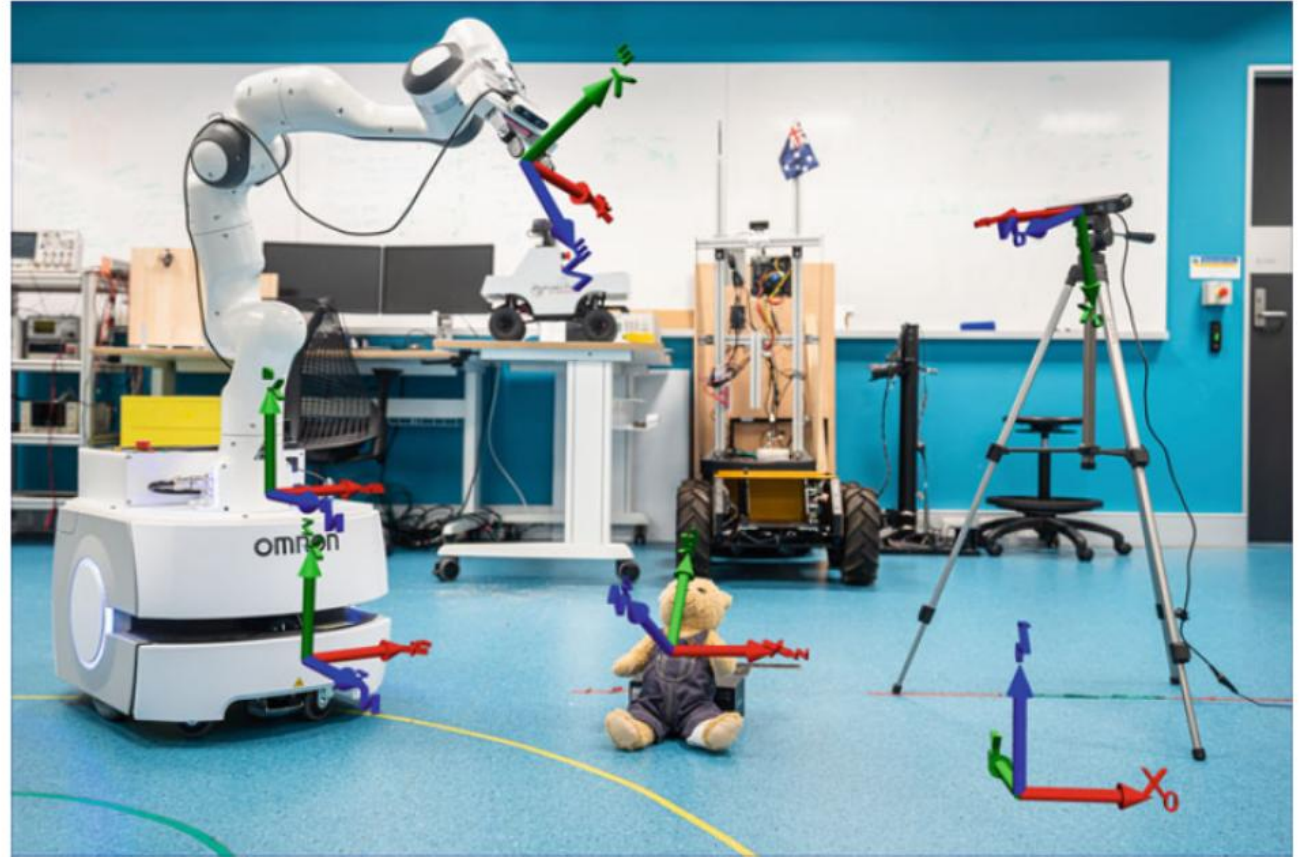
Reference Text

- Robot modeling and control by Spong et al. (Chapter 2)
- Robotics, Vision, and Control by Peter Corke (Chapter 2)



Coordinate Transformation

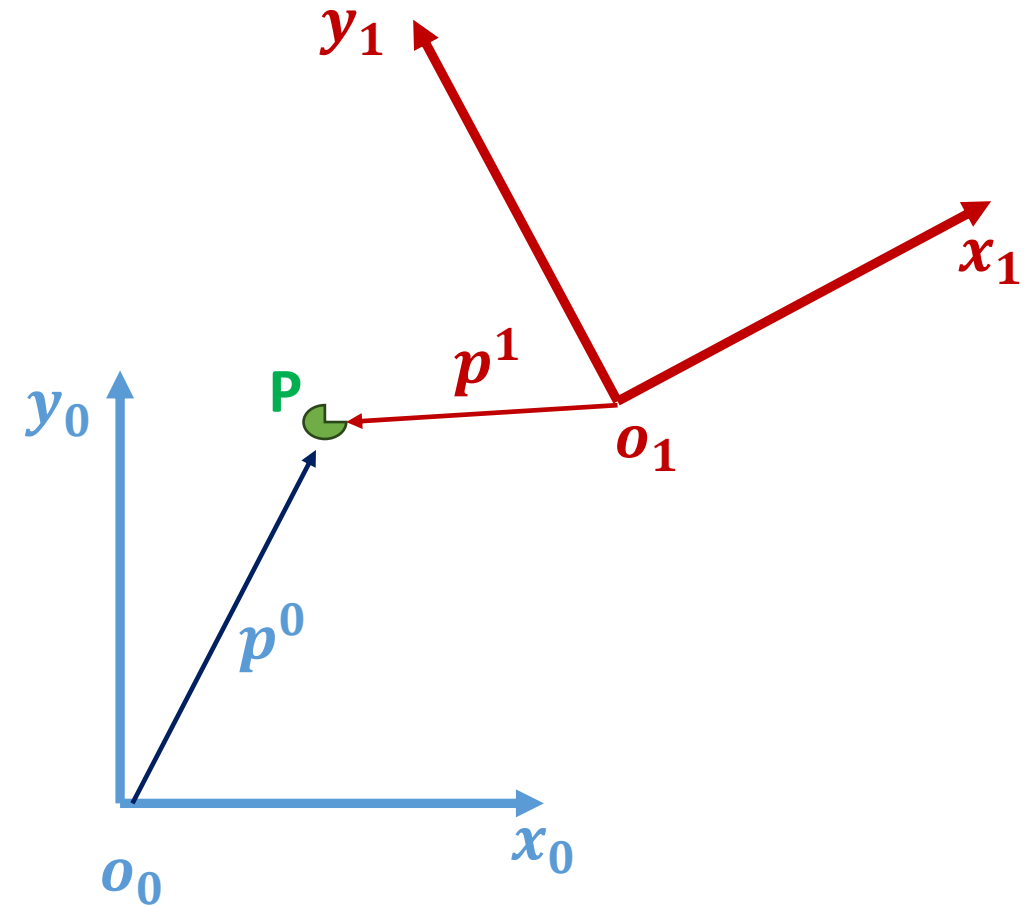
- An important aspect of robot kinematics is establishing various **coordinate systems** to represent the **positions and orientations** of rigid objects and understanding **transformations among these coordinate systems**



■ **Fig. 2.6** Examples of coordinate frames for a real-world scenario. Frames include the mobile robot $\{M\}$, arm base $\{B\}$, arm end effector $\{E\}$, camera $\{C\}$, workpiece $\{P\}$ and the reference frame $\{0\}$ (image by John Skinner and Dorian Tsai)

Representing Positions

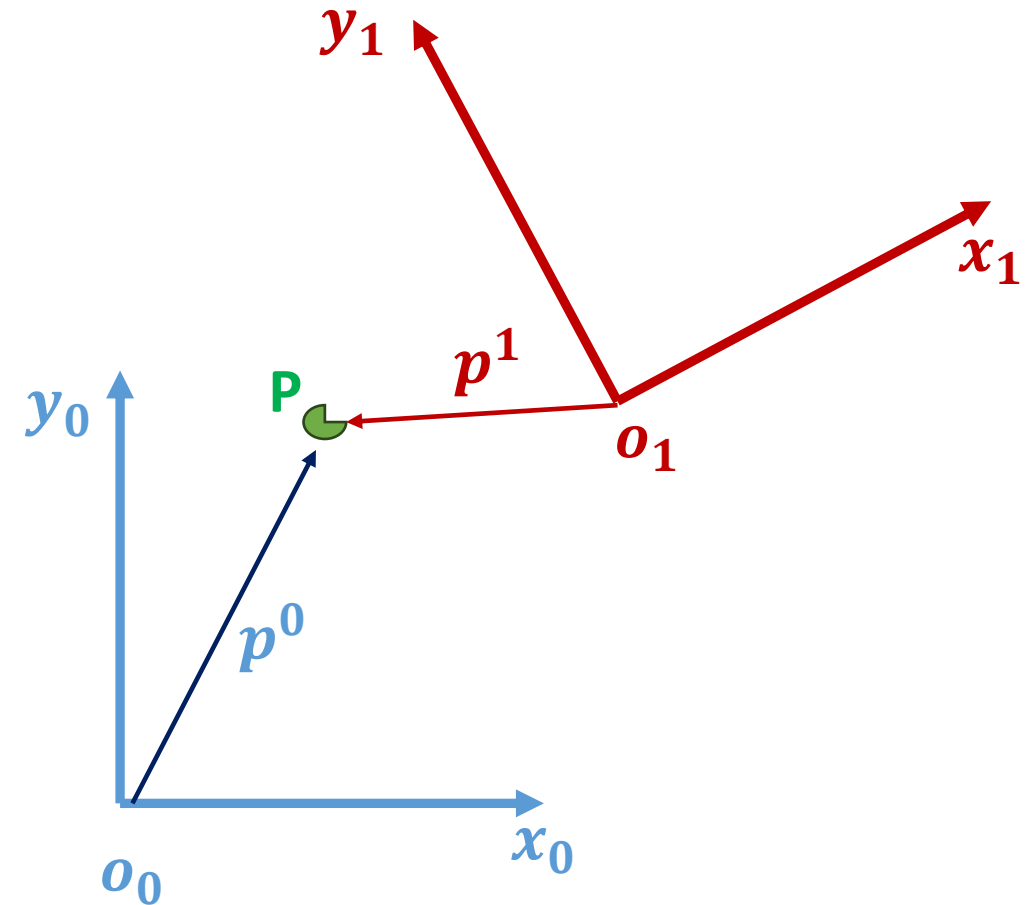
- Two coordinate frames:
 - Frame 0: (o_0, x_0, y_0)
 - Frame 1: (o_1, x_1, y_1)
- An event of interest occurs at point P .



Representing Positions

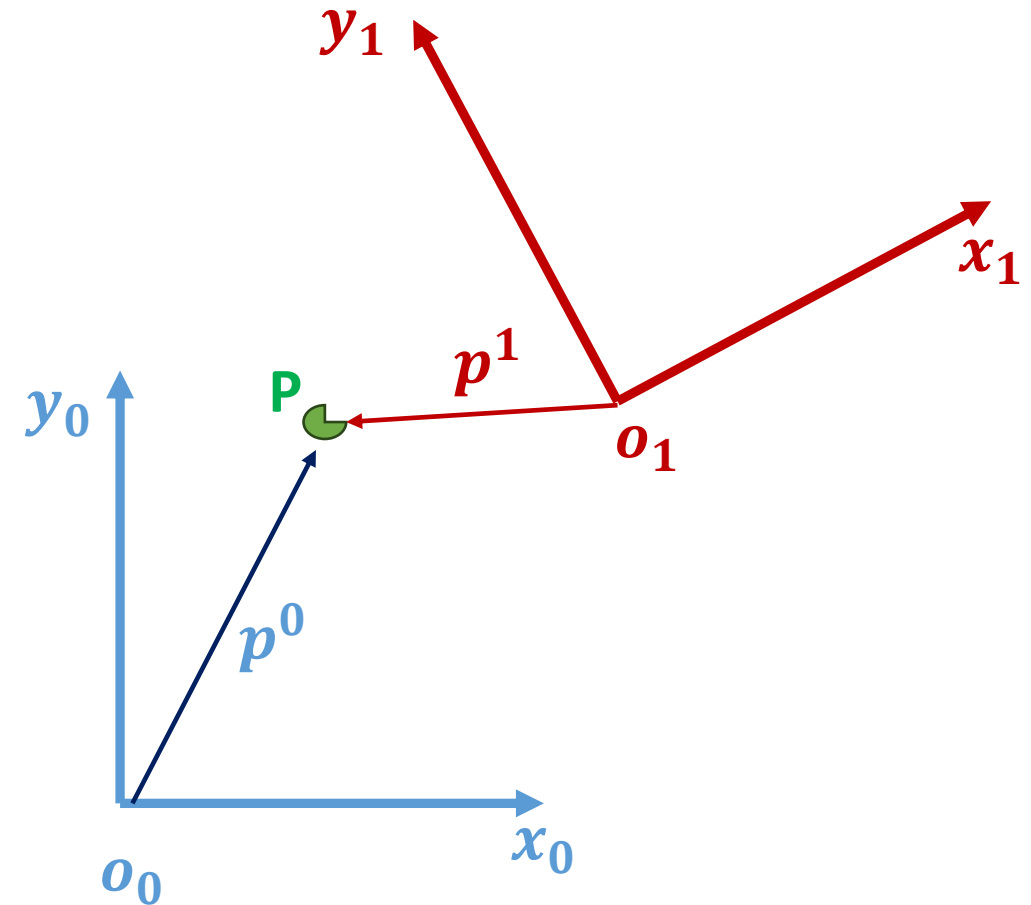
- Two coordinate frames:
 - Frame 0: (o_0, x_0, y_0)
 - Frame 1: (o_1, x_1, y_1)
- An event of interest occurs at point P .
- We can specify coordinates of P with respect to Frame 0 or Frame 1.
- For example:

$$p^0 = (3, 5) \quad p^1 = (-3, 0.5)$$



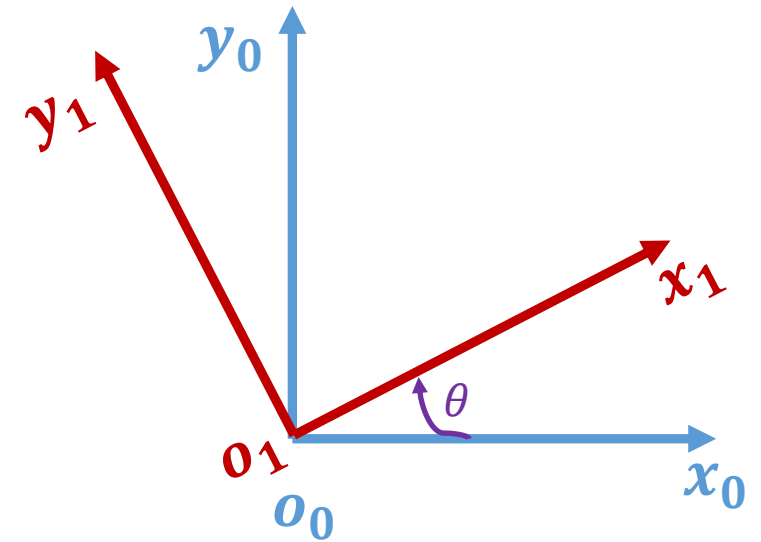
Representing Positions

- Two coordinate frames:
 - Frame 0: (o_0, x_0, y_0)
 - Frame 1: (o_1, x_1, y_1)
- The origins of the frames are also points.
$$o_1^0 = (12, 8) \quad o_0^1 = (-12, -8)$$
- o_1^0 : Coordinates of point o_1 relative to frame 0.
- o_0^1 : Coordinates of point o_0 relative to frame 1.



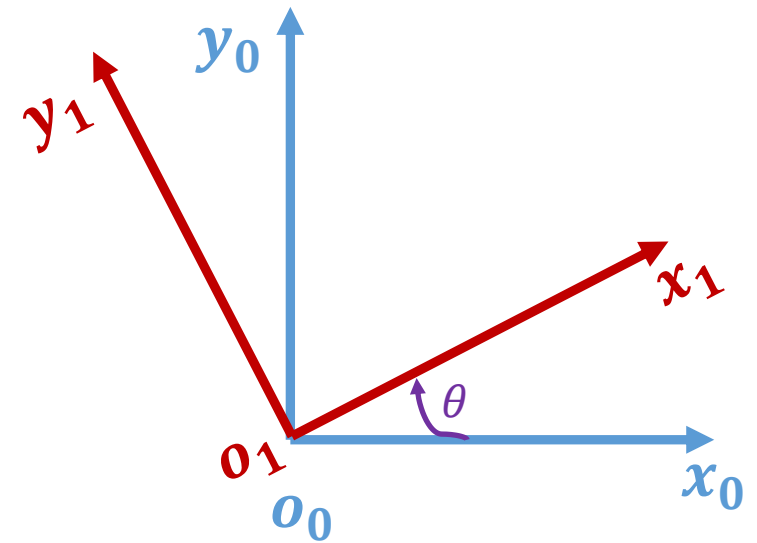
Representing Rotations

- Two coordinate frames:
 - Frame 0: (o_0, x_0, y_0)
 - Frame 1: (o_1, x_1, y_1)
- Frame 1 is obtained by rotation Frame 0 by an angle θ
- How to represent relative orientation of the two frames?



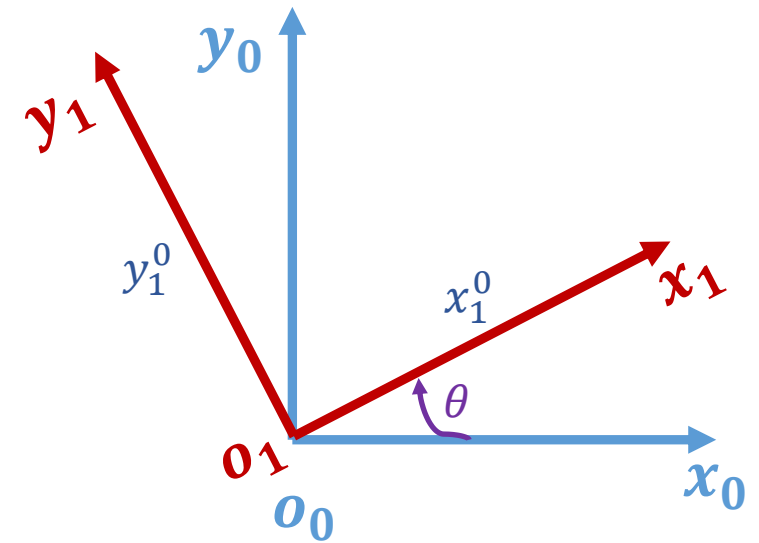
Representing Rotations

- Option 1: Just use rotation angle θ .
- **Issue:** Discontinuity around $\theta = 0$.
- For $\theta = 2\pi - \epsilon$, small changes in orientation can produce large changes in the value of θ .
- A rotation by ϵ , causes θ to “wrap around” to zero.
- Does not scale well to three dimensions.



Representing Rotations

- Option 2: Specify the coordinate vectors of Frame 1 with respect to Frame 0.
- $R_1^0 = [x_1^0 \mid y_1^0]$
- Here
 - x_1^0 : Representation of x_1 with respect to Frame 0: (o_0, x_0, y_0)
 - y_1^0 : Representation of y_1 with respect to Frame 0: (o_0, x_0, y_0)
 - R_1^0 : Rotation matrix

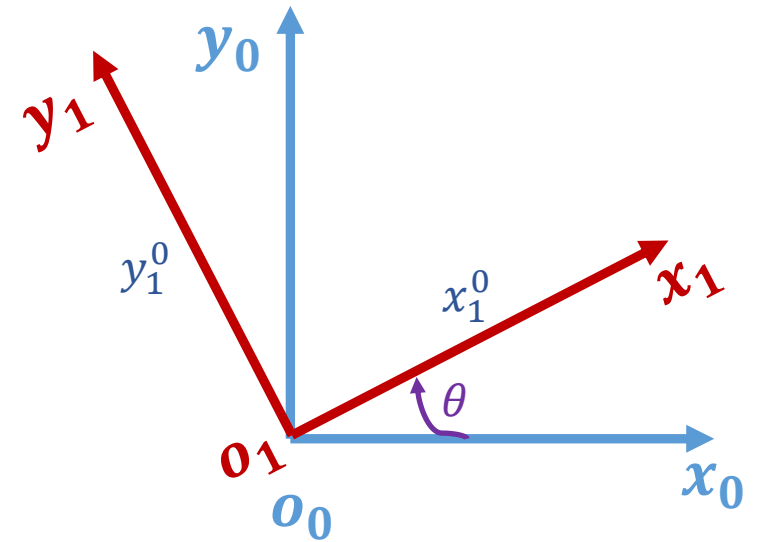


Representing Rotations

- Option 2: Specify the coordinate vectors of Frame 1 with respect to Frame 0.
- $R_1^0 = [x_1^0 \mid y_1^0]$

$$x_1^0 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} \quad y_1^0 = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

$$R_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$



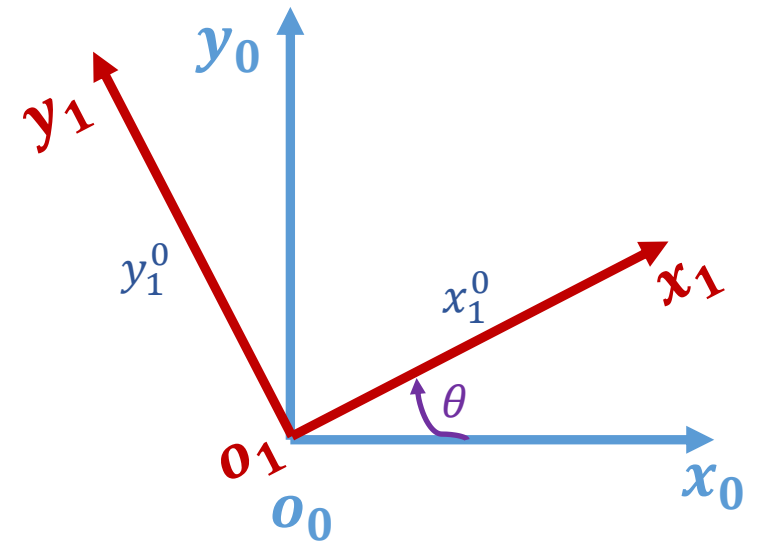
Representing Rotations

- Option 2: Specify the coordinate vectors of Frame 1 with respect to Frame 0.

- $R_1^0 = [x_1^0 \mid y_1^0]$

$$R_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

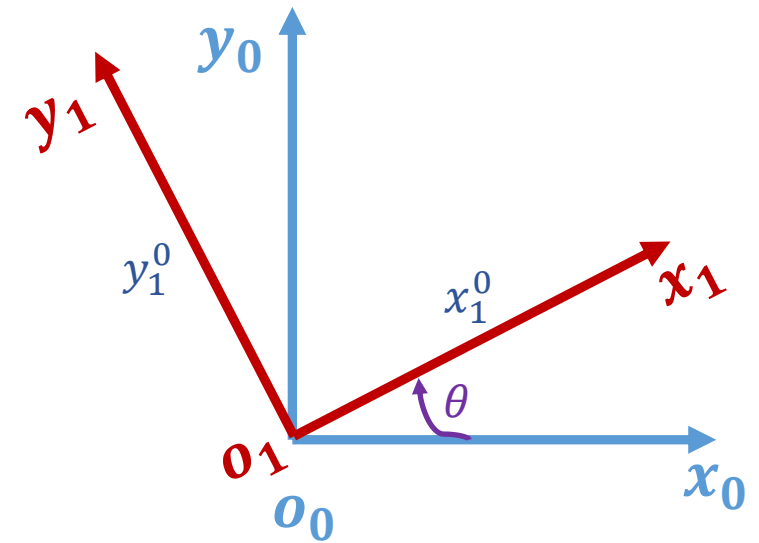
- R_1^0 : Matrix whose column vectors are the coordinates of the unit vectors along the axes of **Frame 1**: (o_1, x_1, y_1) expressed relative to **Frame 0**: (o_0, x_0, y_0)



Representing Rotations

- Build rotation matrix by projecting the axis of **Frame 1**: (o_1, x_1, y_1) onto the coordinate axis of **Frame 0**: (o_0, x_0, y_0)
- The dot product of two unit vectors gives the projection of one onto the other.

$$x_1^0 = \begin{bmatrix} x_1 \cdot x_0 \\ x_1 \cdot y_0 \end{bmatrix} \quad y_1^0 = \begin{bmatrix} y_1 \cdot x_0 \\ y_1 \cdot y_0 \end{bmatrix}$$

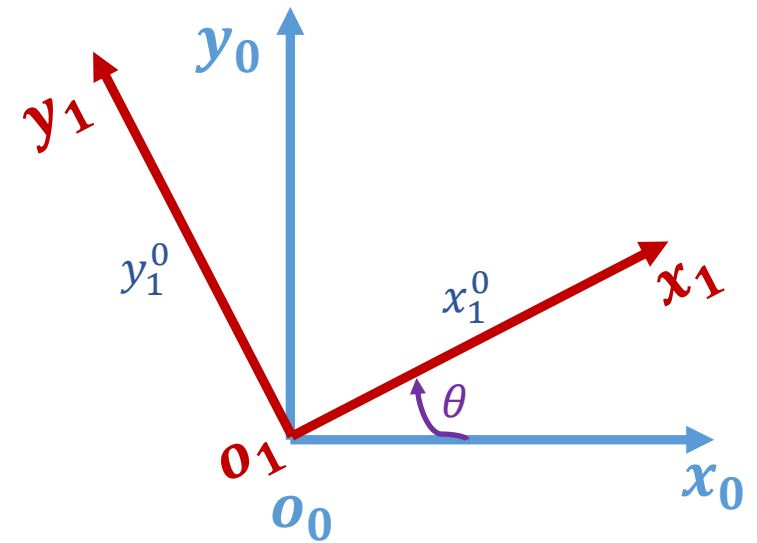


Representing Rotations

- Build rotation matrix by projecting the axis of **Frame 1: (o_1, x_1, y_1)** onto the coordinate axis of **Frame 0: (o_0, x_0, y_0)**

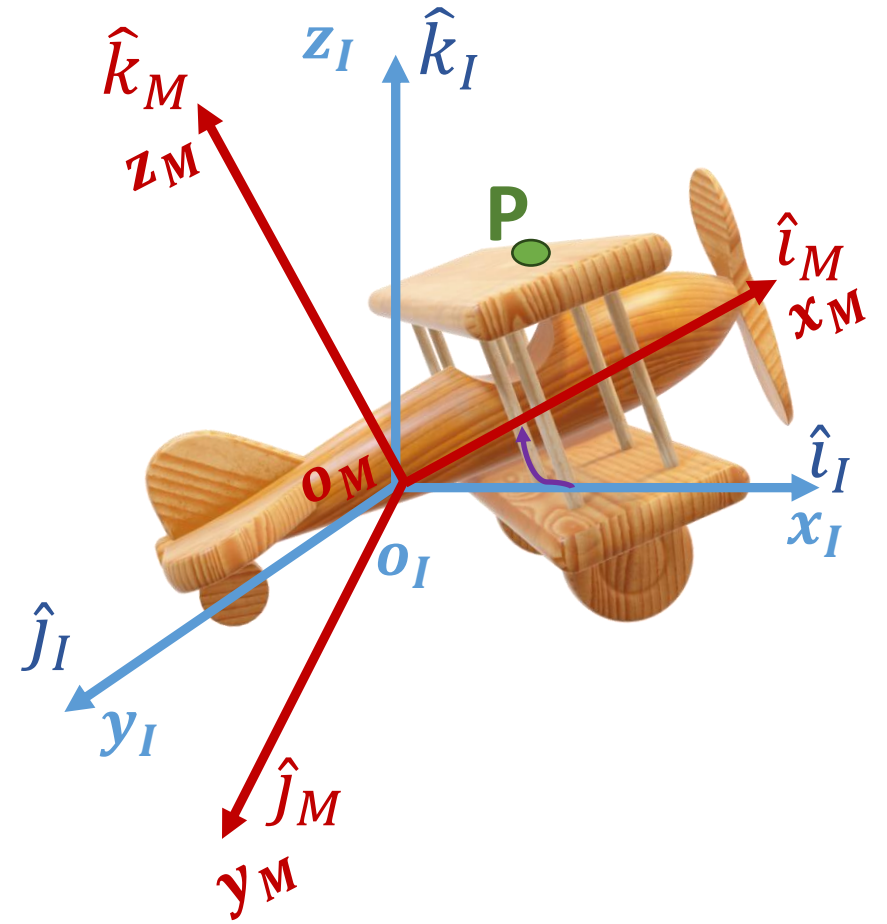
$$R_1^0 = \begin{bmatrix} x_1 \cdot x_0 & y_1 \cdot x_0 \\ x_1 \cdot y_0 & y_1 \cdot y_0 \end{bmatrix}$$

- The columns of R_1^0 specify the direction cosines of the coordinate axis of **(o_1, x_1, y_1)** relative to the coordinate axis of **(o_0, x_0, y_0)**



Rotations in Three Dimensions

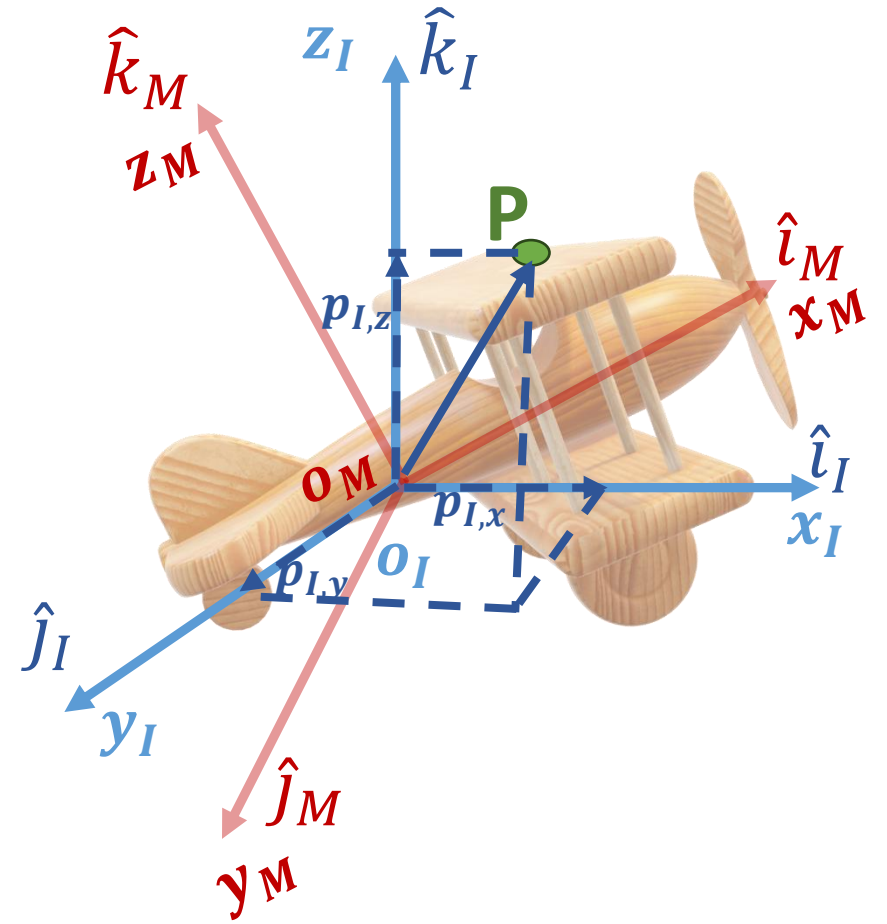
- Let (x_I, y_I, z_I) represent the inertial frame of reference with origin o_I and unit vectors $\hat{i}_I, \hat{j}_I$, and $\hat{k}_I$.
- Let (x_M, y_M, z_M) represent the mobile frame of reference with origin o_M and unit vectors $\hat{i}_M, \hat{j}_M$, and $\hat{k}_M$.
- Consider an arbitrary point P in the three-dimensional frame.



Rotations in Three Dimensions

- Point P can be described with respect to the inertial as well as robot frame of reference.
- With respect to inertial frame

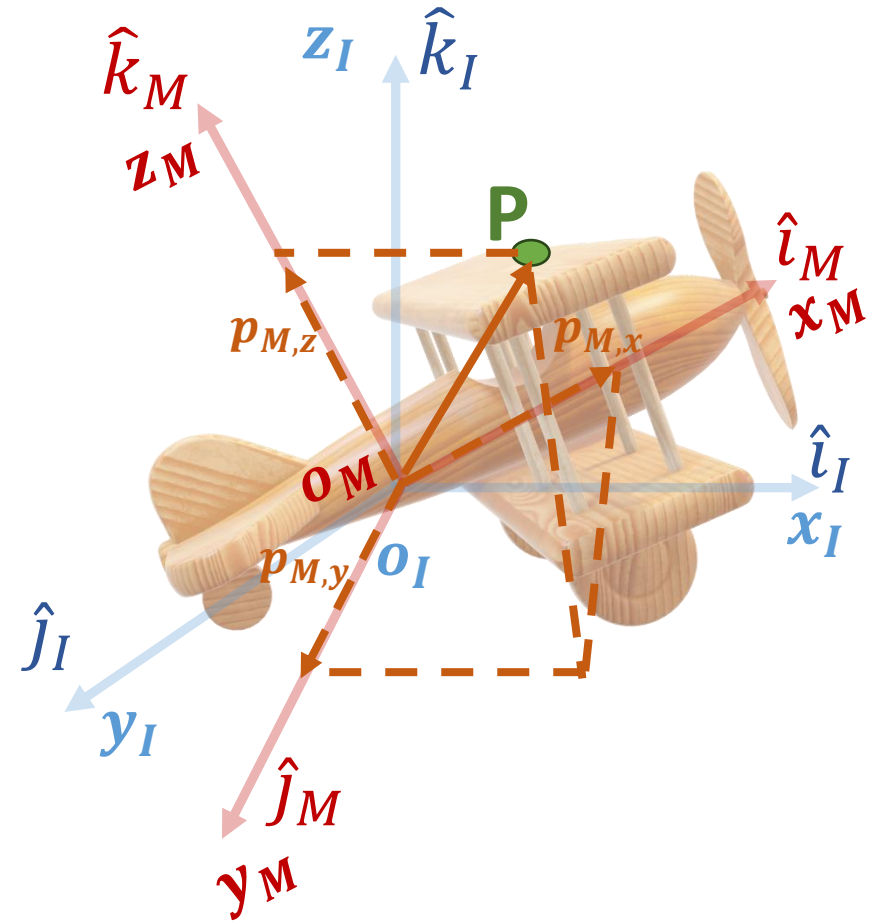
$$P_I = p_{I,x} \hat{i}_I + p_{I,y} \hat{j}_I + p_{I,z} \hat{k}_I$$



Rotations in Three Dimensions

- Point P can be described with respect to the inertial as well as robot frame of reference.
- With respect to **robot frame**

$$P_M = p_{M,x} \hat{i}_M + p_{M,y} \hat{j}_M + p_{M,z} \hat{k}_M$$



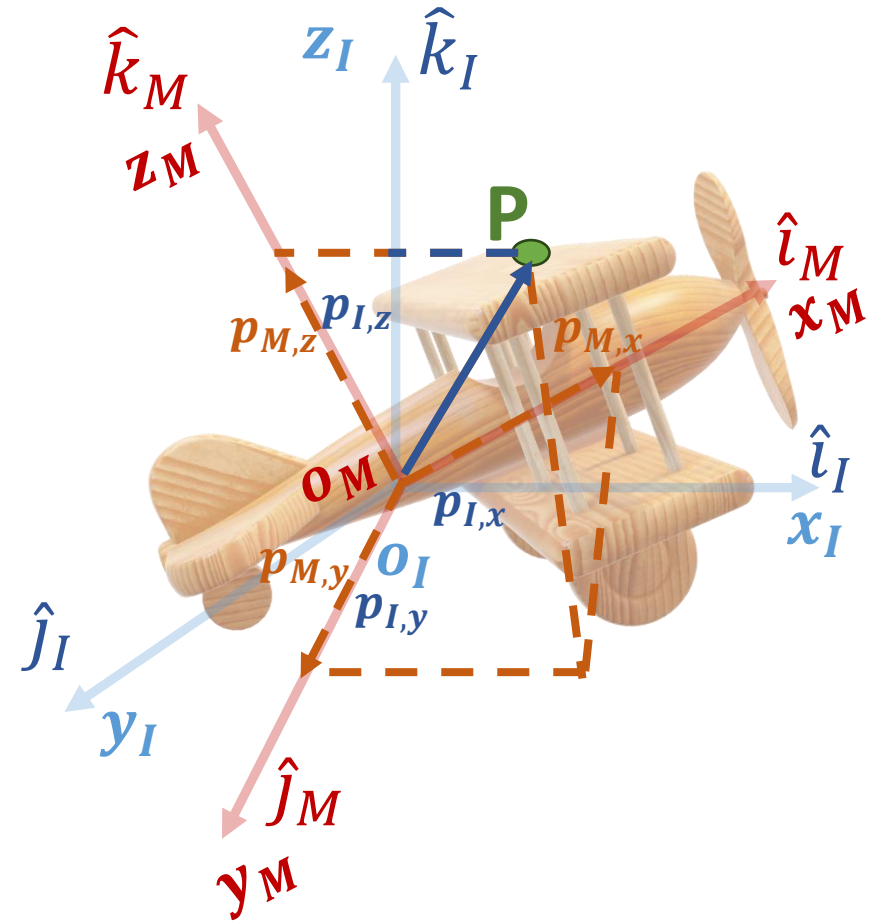
Rotations in Three Dimensions

- Point P can be described with respect to the inertial as well as robot frame of reference.
- With respect to inertial frame

$$P_I = p_{I,x} \hat{i}_I + p_{I,y} \hat{j}_I + p_{I,z} \hat{k}_I$$

- With respect to robot frame

$$P_M = p_{M,x} \hat{i}_M + p_{M,y} \hat{j}_M + p_{M,z} \hat{k}_M$$



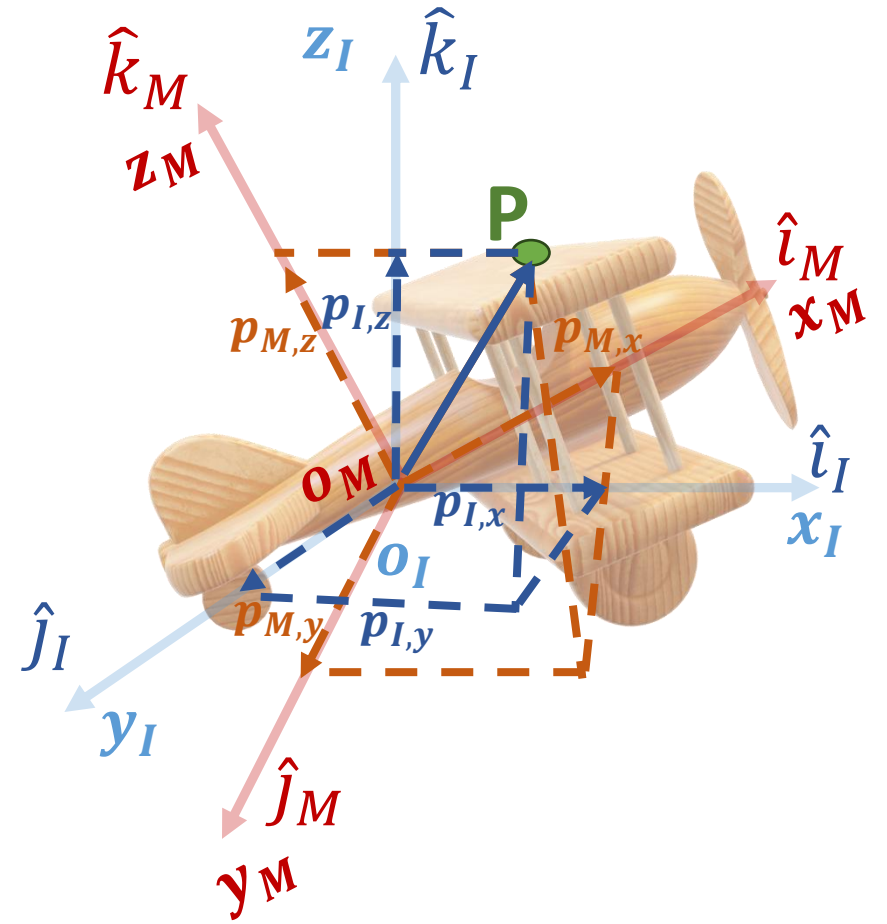
Rotations in Three Dimensions

- Since P_I and P_M represent the **same point**, but in **different frames**, the relationship between the components in the two representations can be obtained as follows:

$$p_{I,x} = P_I \cdot \hat{i}_I = P_M \cdot \hat{i}_I$$

- Based on the above expression

$$p_{I,x} = p_{M,x} \hat{i}_M \cdot \hat{i}_I + p_{M,y} \hat{j}_M \cdot \hat{i}_I + p_{M,z} \hat{k}_M \cdot \hat{i}_I$$



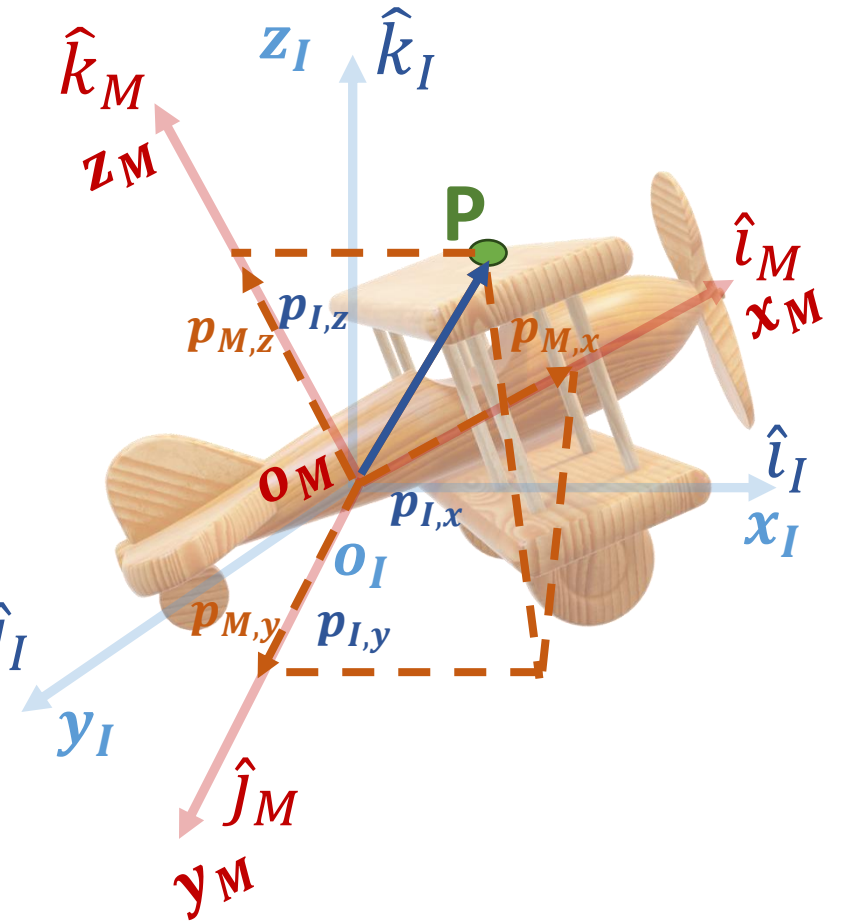
Rotations in Three Dimensions

- We can derive similar representations for $p_{I,y}$ and $p_{I,z}$.

$$p_{I,x} = p_{M,x} \hat{i}_M \cdot \hat{i}_I + p_{M,y} \hat{j}_M \cdot \hat{i}_I + p_{M,z} \hat{k}_M \cdot \hat{i}_I$$

$$p_{I,y} = p_{M,x} \hat{i}_M \cdot \hat{j}_I + p_{M,y} \hat{j}_M \cdot \hat{j}_I + p_{M,z} \hat{k}_M \cdot \hat{j}_I$$

$$p_{I,z} = p_{M,x} \hat{i}_M \cdot \hat{k}_I + p_{M,y} \hat{j}_M \cdot \hat{k}_I + p_{M,z} \hat{k}_M \cdot \hat{k}_I$$

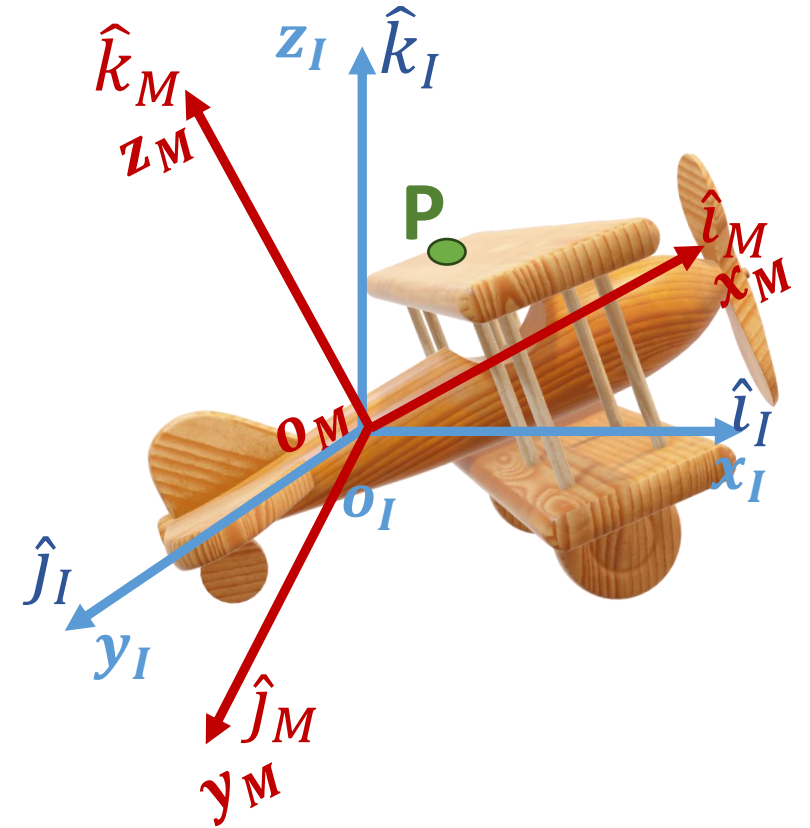


Rotations in Three Dimensions

- We can express these relationships in matrix form as well.

$$\begin{bmatrix} p_{I,x} \\ p_{I,y} \\ p_{I,z} \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{i}_M \cdot \hat{i}_I & \hat{j}_M \cdot \hat{i}_I & \hat{k}_M \cdot \hat{i}_I \\ \hat{i}_M \cdot \hat{j}_I & \hat{j}_M \cdot \hat{j}_I & \hat{k}_M \cdot \hat{j}_I \\ \hat{i}_M \cdot \hat{k}_I & \hat{j}_M \cdot \hat{k}_I & \hat{k}_M \cdot \hat{k}_I \end{bmatrix}}_{R_M^I} \begin{bmatrix} p_{M,x} \\ p_{M,y} \\ p_{M,z} \end{bmatrix}$$

$$P_I = R_M^I P_M$$

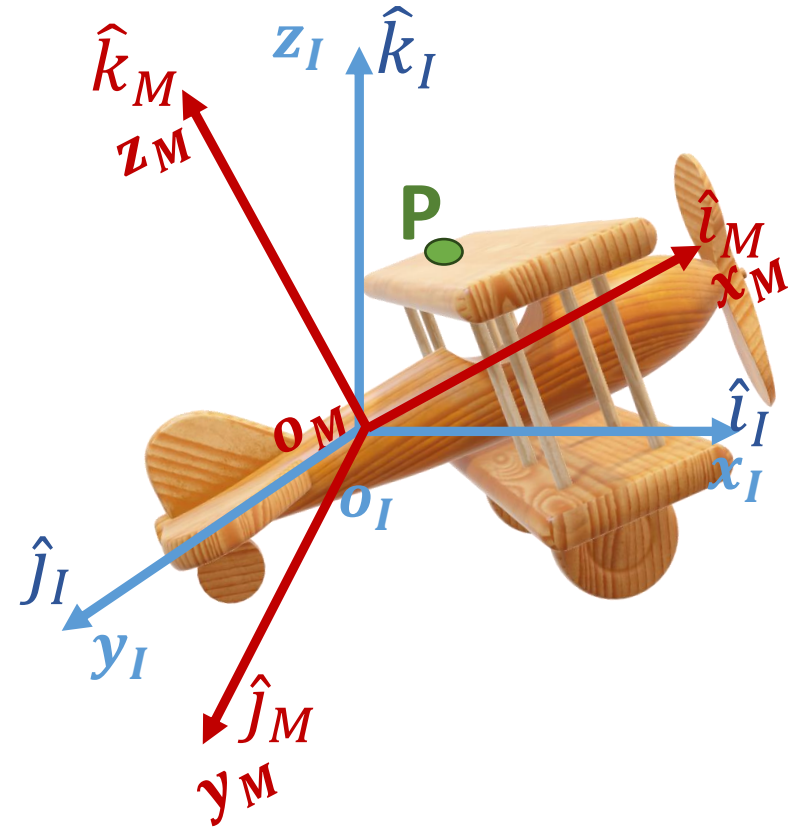


Rotations in Three Dimensions

- We can express these relationships in matrix form as well.

$$P_I = R_M^I P_M$$

- The matrix R_M^I is the rotation matrix, which represents the transformation from the coordinates of the point P with respect to the robot frame (x_M, y_M, z_M) to the coordinates with respect to inertial frame (x_I, y_I, z_I) .

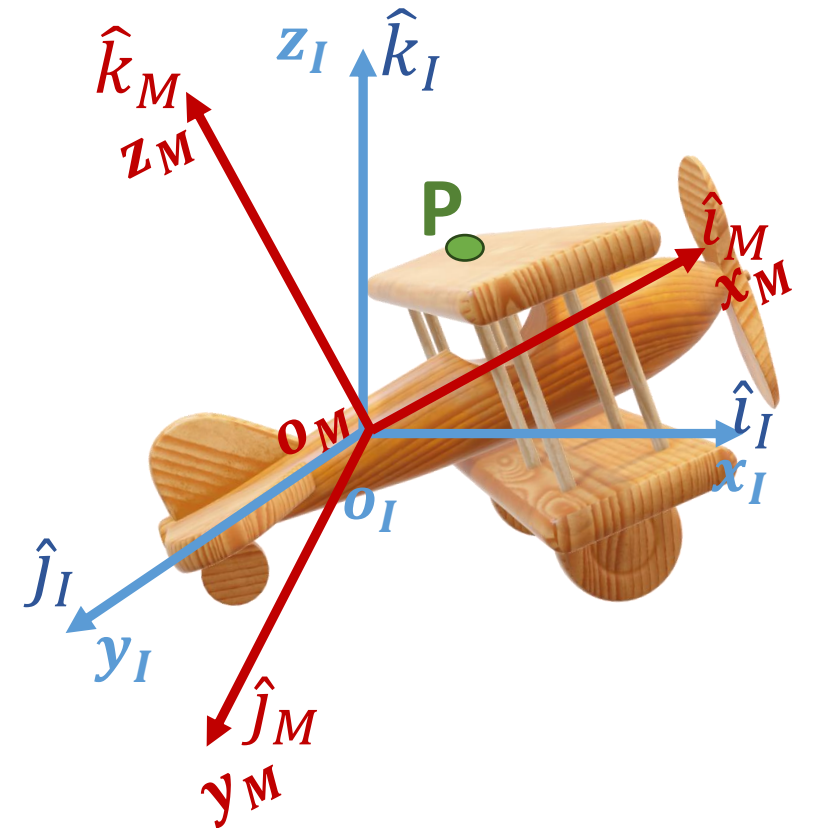


Rotations in Three Dimensions

- The rotation matrix is defined in terms of dot products of unit vectors; it can be represented as direction cosines as follows:

$$P_I = R_M^I P_M$$

$$R_M^I = \begin{bmatrix} \cos(\theta_{i_M, i_I}) & \cos(\theta_{j_M, i_I}) & \cos(\theta_{k_M, i_I}) \\ \cos(\theta_{i_M, j_I}) & \cos(\theta_{j_M, j_I}) & \cos(\theta_{k_M, j_I}) \\ \cos(\theta_{i_M, k_I}) & \cos(\theta_{j_M, k_I}) & \cos(\theta_{k_M, k_I}) \end{bmatrix}$$



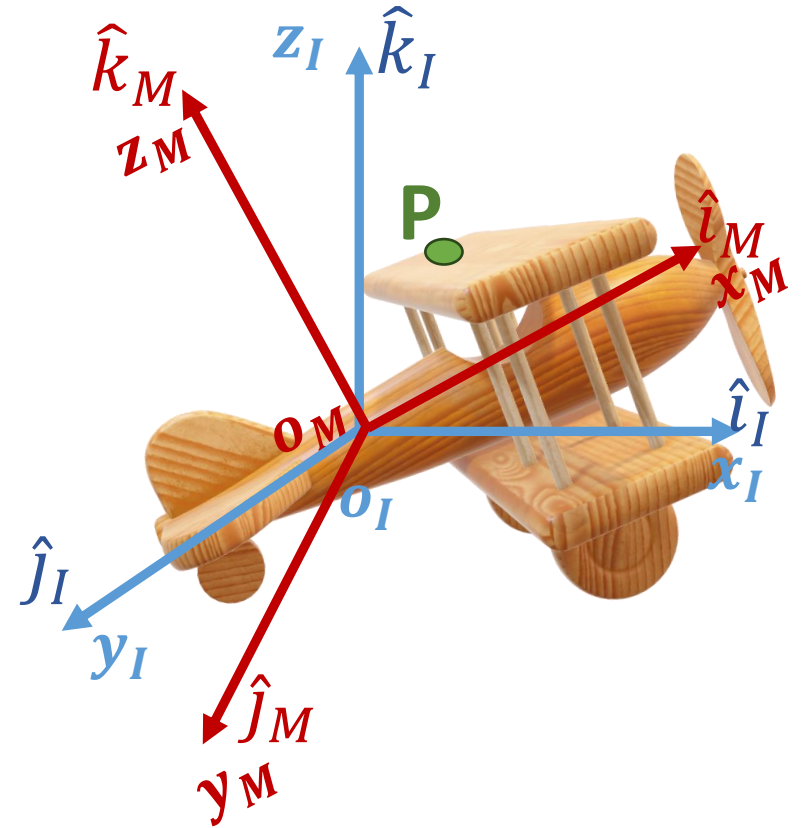
Rotations in Three Dimensions

- Similarly, we can express these relationships in the opposite direction as well.

$$p_{M,x} = p_{I,x} \hat{i}_I \cdot \hat{i}_M + p_{I,y} \hat{j}_I \cdot \hat{i}_M + p_{I,z} \hat{k}_I \cdot \hat{i}_M$$

$$p_{M,y} = p_{I,x} \hat{i}_I \cdot \hat{j}_M + p_{I,y} \hat{j}_I \cdot \hat{j}_M + p_{I,z} \hat{k}_I \cdot \hat{j}_M$$

$$p_{M,z} = p_{I,x} \hat{i}_I \cdot \hat{k}_M + p_{I,y} \hat{j}_I \cdot \hat{k}_M + p_{I,z} \hat{k}_I \cdot \hat{k}_M$$

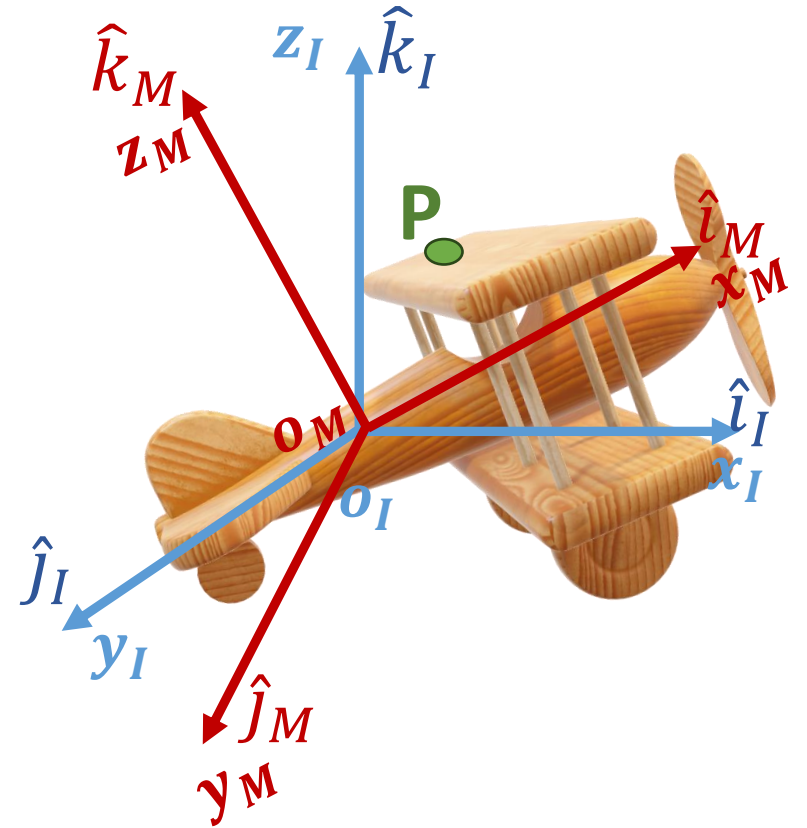


Rotations in Three Dimensions

- Similarly, we can express these relationships in the opposite direction as well.

$$P_M = R_I^M P_I$$

$$\begin{bmatrix} p_{M,x} \\ p_{M,y} \\ p_{M,z} \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{i}_I \cdot \hat{i}_M & \hat{j}_I \cdot \hat{i}_M & \hat{k}_I \cdot \hat{i}_M \\ \hat{i}_I \cdot \hat{j}_M & \hat{j}_I \cdot \hat{j}_M & \hat{k}_I \cdot \hat{j}_M \\ \hat{i}_I \cdot \hat{k}_M & \hat{j}_I \cdot \hat{k}_M & \hat{k}_I \cdot \hat{k}_M \end{bmatrix}}_{R_I^M} \begin{bmatrix} p_{I,x} \\ p_{I,y} \\ p_{I,z} \end{bmatrix}$$



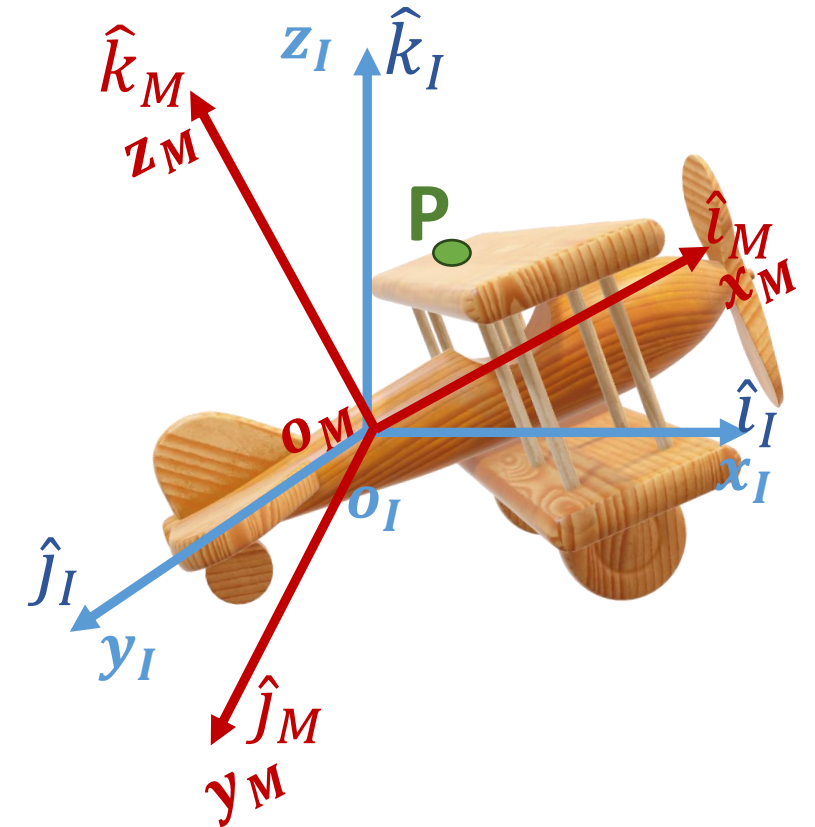
Rotations in Three Dimensions

- Similarly, we can express these relationships in the opposite direction as well.

$$P_M = R_I^M P_I$$

$$\begin{bmatrix} p_{M,x} \\ p_{M,y} \\ p_{M,z} \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{i}_I \cdot \hat{i}_M & \hat{j}_I \cdot \hat{i}_M & \hat{k}_I \cdot \hat{i}_M \\ \hat{i}_I \cdot \hat{j}_M & \hat{j}_I \cdot \hat{j}_M & \hat{k}_I \cdot \hat{j}_M \\ \hat{i}_I \cdot \hat{k}_M & \hat{j}_I \cdot \hat{k}_M & \hat{k}_I \cdot \hat{k}_M \end{bmatrix}}_{R_I^M} \begin{bmatrix} p_{I,x} \\ p_{I,y} \\ p_{I,z} \end{bmatrix}$$

$$R_I^M = \begin{bmatrix} \cos(\theta_{i_M, i_I}) & \cos(\theta_{i_M, j_I}) & \cos(\theta_{i_M, k_I}) \\ \cos(\theta_{j_M, i_I}) & \cos(\theta_{j_M, j_I}) & \cos(\theta_{j_M, k_I}) \\ \cos(\theta_{k_M, i_I}) & \cos(\theta_{k_M, j_I}) & \cos(\theta_{k_M, k_I}) \end{bmatrix}$$



Rotations in Three Dimensions

- The two rotation matrices R_M^I and R_I^M are inverse of each other.

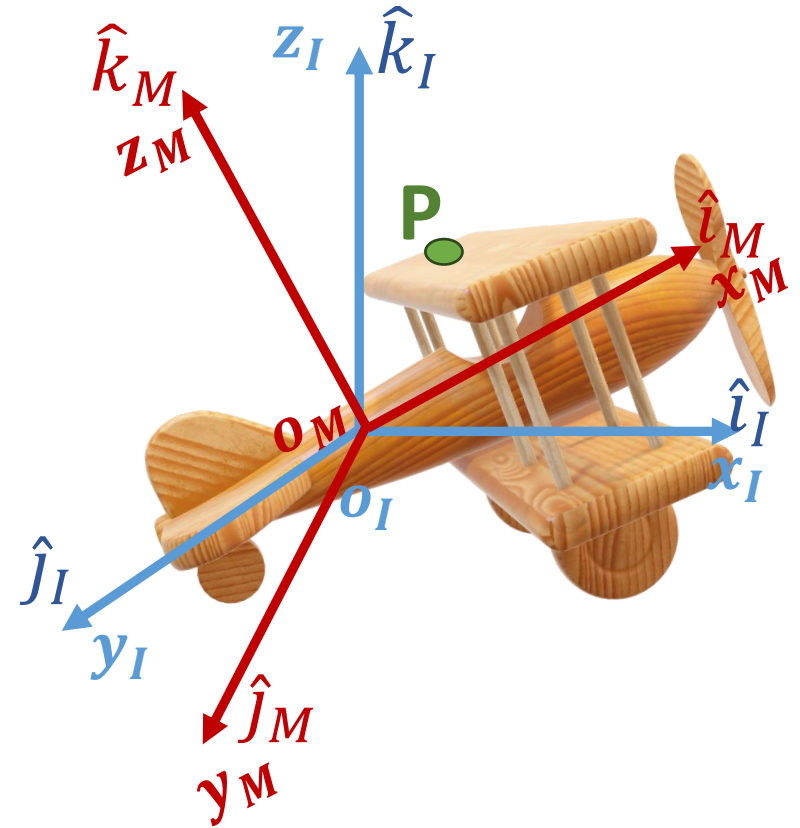
$$R_M^I = (R_I^M)^{-1}$$

- Since inner product is commutative,

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i},$$

We can easily verify that

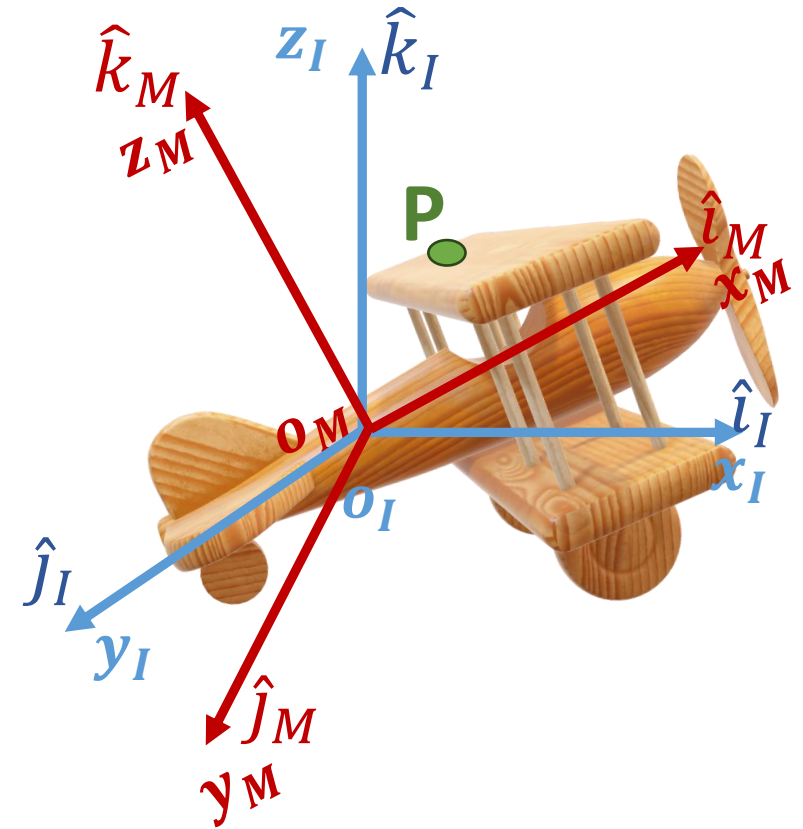
$$R_M^I = (R_I^M)^{-1} = (R_I^M)^T$$



Orthogonal Matrix

$$R_M^I = (R_I^M)^{-1} = (R_I^M)^T$$

- A matrix is called an **Orthogonal Matrix** if its inverse is equal to its transpose.

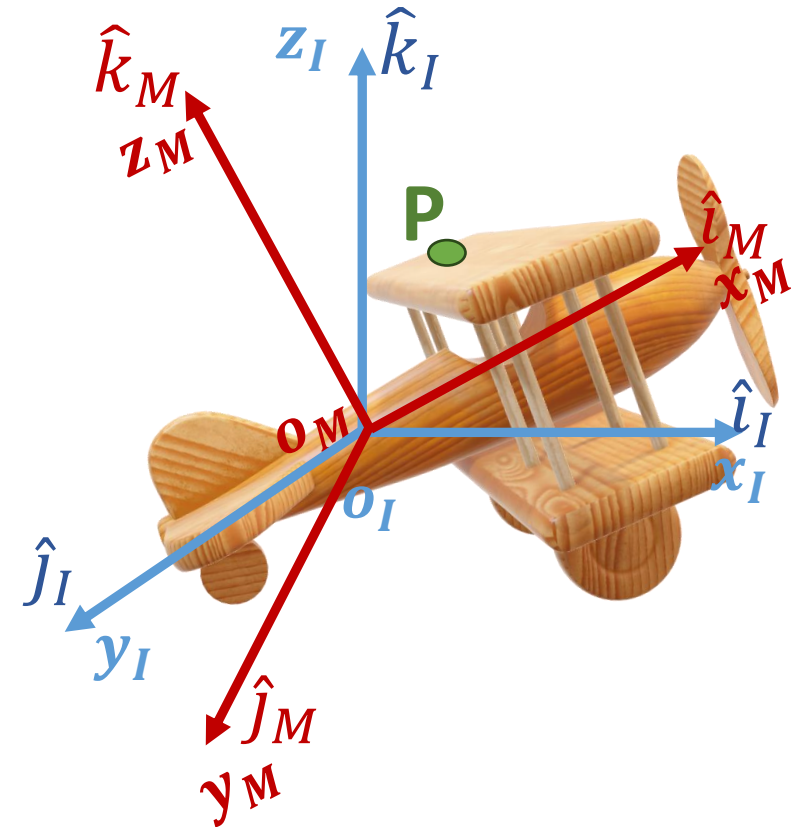


Orthogonal Matrix

$$R_{\textcolor{red}{M}}^{\textcolor{blue}{I}} = (R_{\textcolor{blue}{I}}^{\textcolor{red}{M}})^{-1} = (R_{\textcolor{blue}{I}}^{\textcolor{red}{M}})^T$$

- A matrix is called an **Orthogonal Matrix** if its **inverse is equal to its transpose**.
- The column vectors of $R_{\textcolor{red}{M}}^{\textcolor{blue}{I}}$ are unit length and orthogonal.

$$R_M^I = \begin{bmatrix} \cos(\theta_{i_M, i_I}) & \cos(\theta_{j_M, i_I}) & \cos(\theta_{k_M, i_I}) \\ \cos(\theta_{i_M, j_I}) & \cos(\theta_{j_M, j_I}) & \cos(\theta_{k_M, j_I}) \\ \cos(\theta_{i_M, k_I}) & \cos(\theta_{j_M, k_I}) & \cos(\theta_{k_M, k_I}) \end{bmatrix}$$

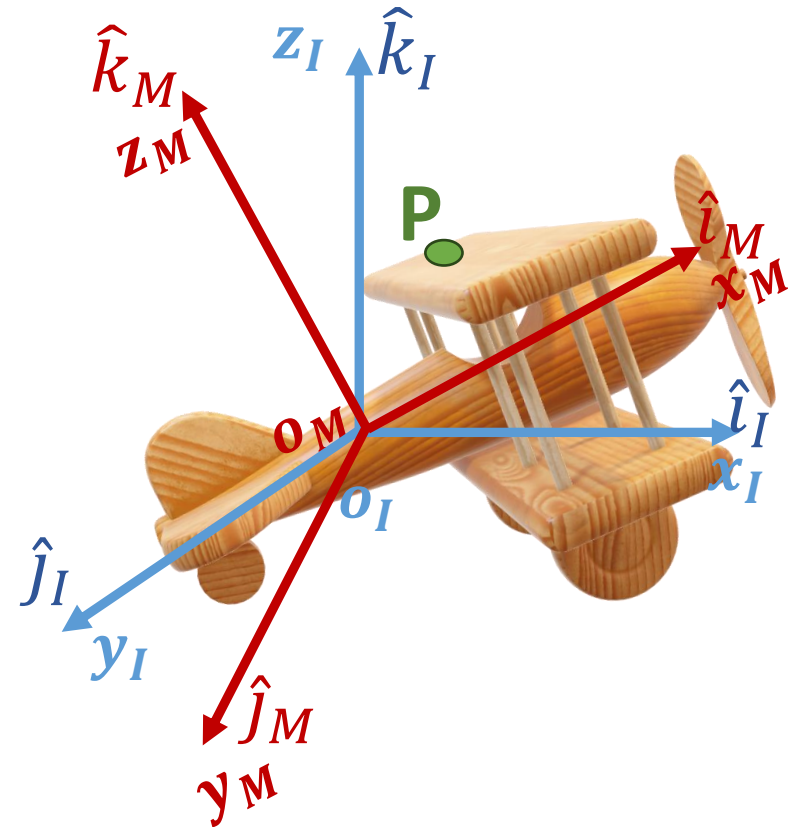


Orthogonal Matrix

$$R_M^I = (R_I^M)^{-1} = (R_I^M)^T$$

- A matrix is called an **Orthogonal Matrix** if its **inverse is equal to its transpose**.
- The column vectors of R_M^I are unit length and orthogonal.
- The rotation matrix R_M^I also has the property

$$\det(R_M^I) = \pm 1$$



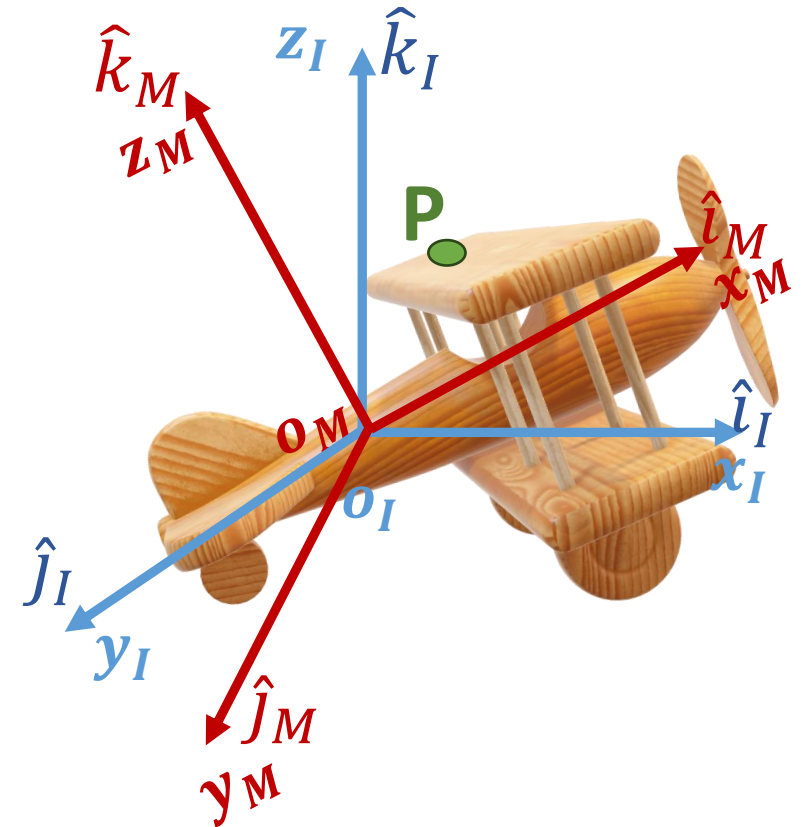
Orthogonal Matrix

$$R_M^I = (R_I^M)^{-1} = (R_I^M)^T$$

- A matrix is called an **Orthogonal Matrix** if its **inverse is equal to its transpose**.
- The column vectors of R_M^I are unit length and orthogonal.
- The rotation matrix R_M^I also has the property

$$\det(R_M^I) = \pm 1$$

- For a right-handed coordinate system
 $\det(R_M^I) = +1$



Orthogonal Matrix

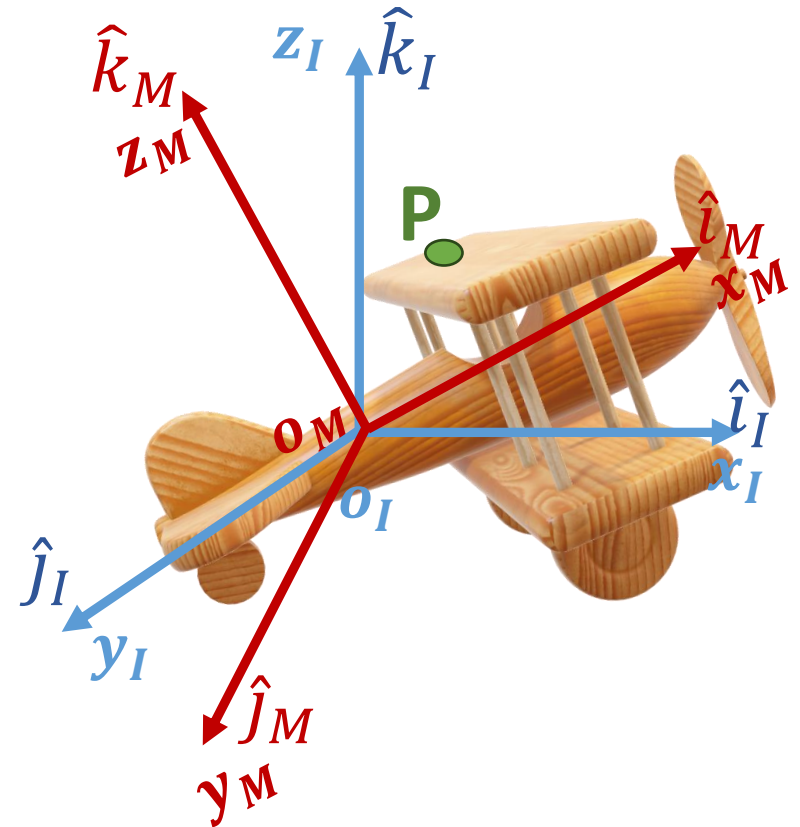
- A matrix is called an **Orthogonal Matrix** if its **inverse is equal to its transpose**.
- The column vectors of R_M^I are unit length and orthogonal.
- The rotation matrix R_M^I also has the property

$$\det(R_M^I) = \pm 1$$

- For a right-handed coordinate system

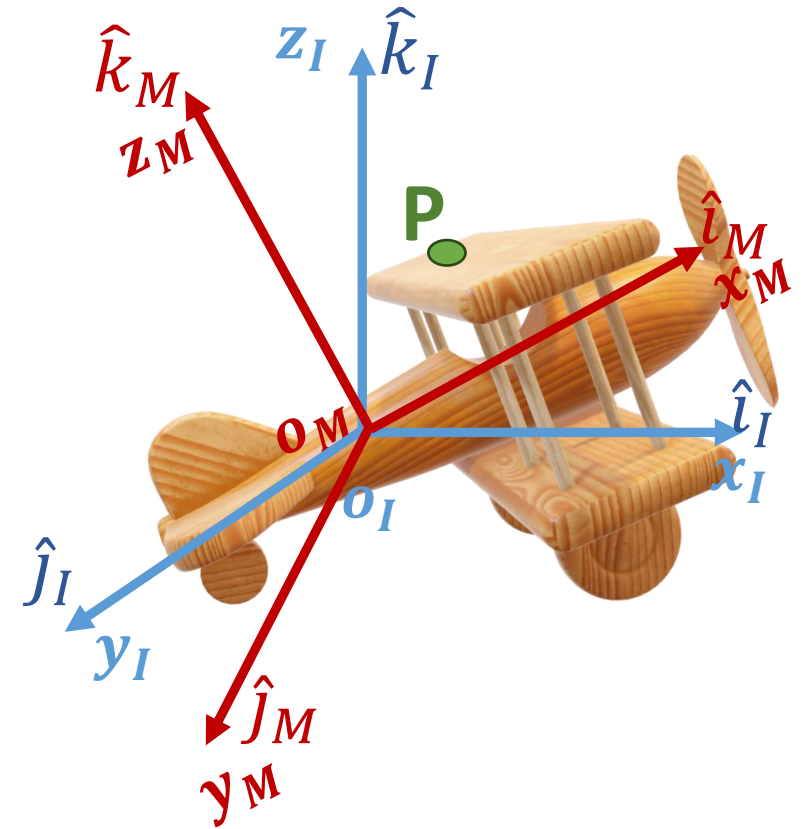
$$\det(R_M^I) = +1$$

- Set of all **3×3 rotation matrices** form a vector space represented by **$SO(3)$**



Example

- Suppose the **robot frame** (x_M, y_M, z_M) is obtained by a rotation of **inertial frame** (x_I, y_I, z_I) by an **angle** θ about z_I axis.

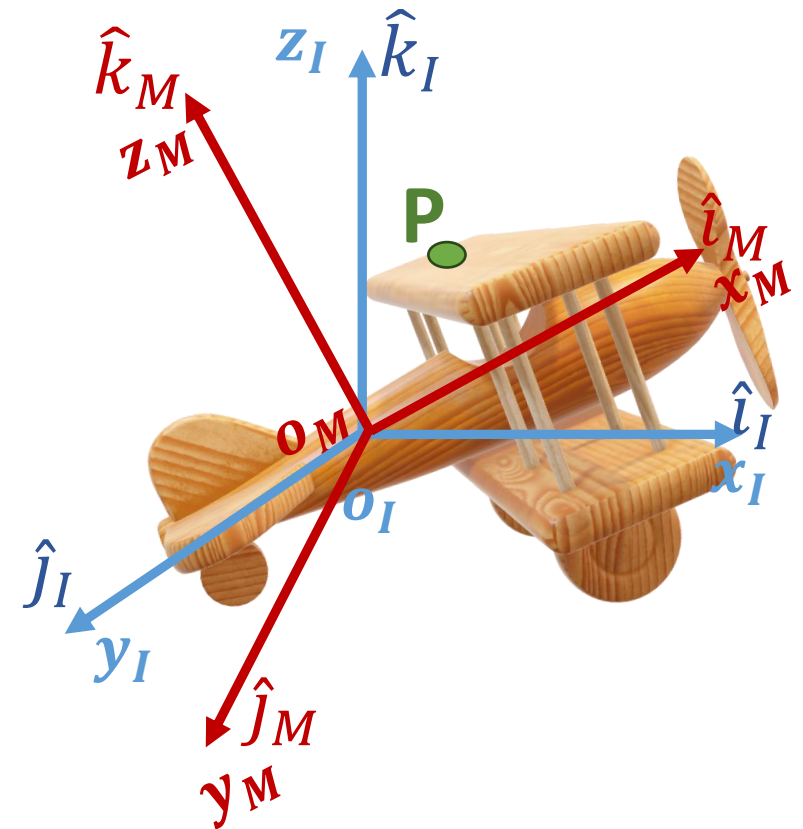


Example

- Suppose the **robot frame** (x_M, y_M, z_M) is obtained by a rotation of **inertial frame** (x_I, y_I, z_I) by an **angle** θ about z_I axis.

$$R_M^I = \begin{bmatrix} \cos(\theta_{i_M, i_I}) & \cos(\theta_{j_M, i_I}) & \cos(\theta_{k_M, i_I}) \\ \cos(\theta_{i_M, j_I}) & \cos(\theta_{j_M, j_I}) & \cos(\theta_{k_M, j_I}) \\ \cos(\theta_{i_M, k_I}) & \cos(\theta_{j_M, k_I}) & \cos(\theta_{k_M, k_I}) \end{bmatrix}$$

- Then,
 - $\hat{i}_I \cdot \hat{i}_M = \cos(\theta)$
 - $\hat{j}_M \cdot \hat{i}_I = \cos\left(\frac{\pi}{2}, \theta\right) = -\sin(\theta)$
 - $\hat{j}_I \cdot \hat{j}_M = \cos(\theta)$
 - $\hat{i}_M \cdot \hat{j}_I = \cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$

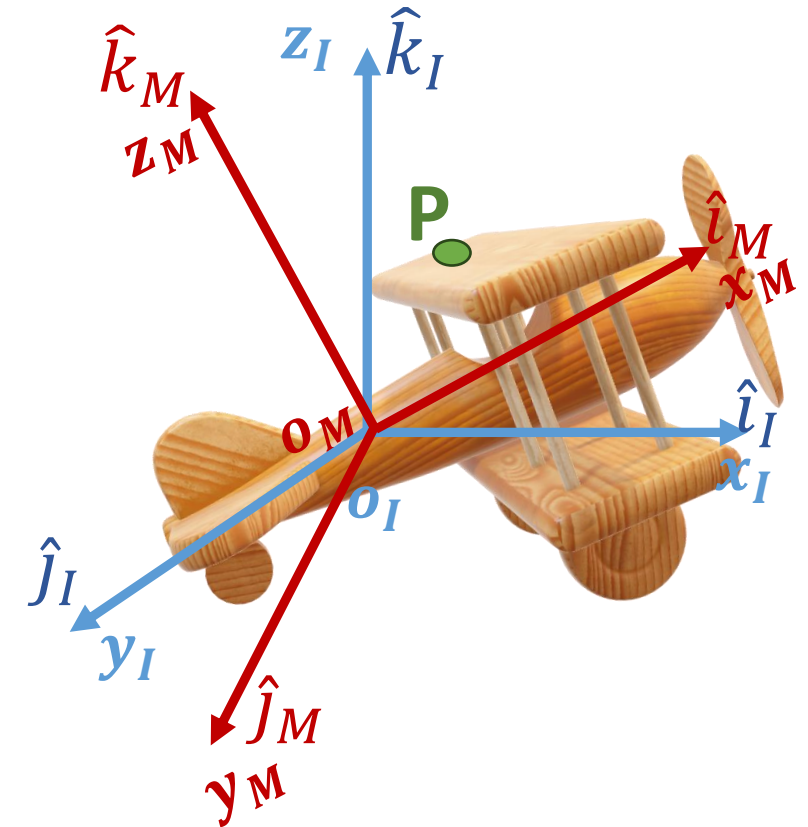


Example

- Suppose the **robot frame** (x_M, y_M, z_M) is obtained by a rotation of **inertial frame** (x_I, y_I, z_I) by an **angle** θ about z_I axis.

$$R_M^I = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

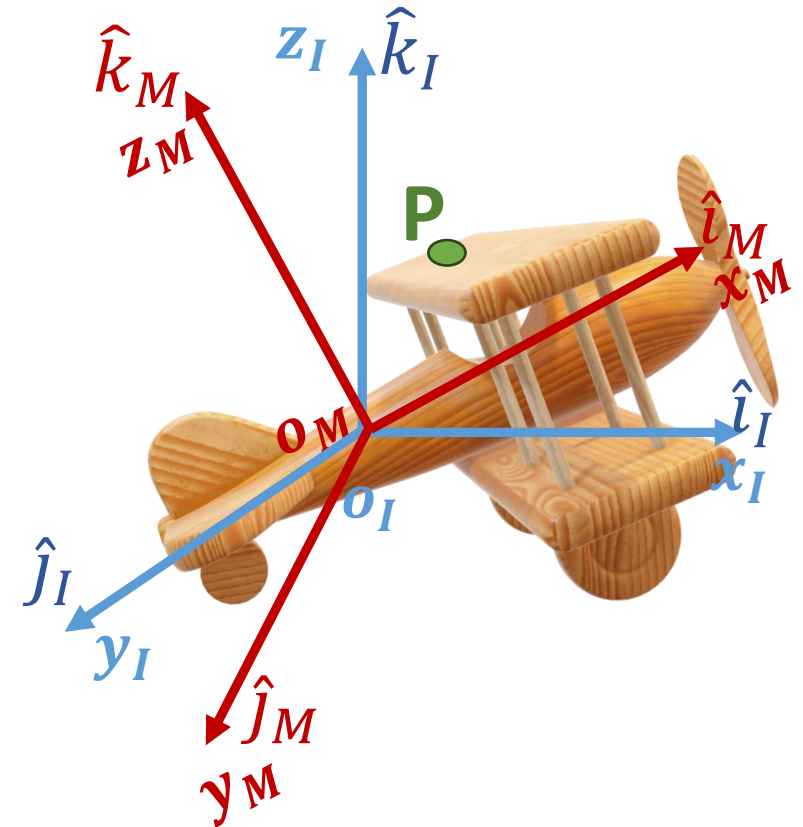
- The above transformation is called a basic rotation about the z-axis.
- A more descriptive notation for this transformation is $R_{z,\theta}$



Example

$$R_M^I = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- A more descriptive notation for this transformation is $R_{z,\theta}$
- This transformation matrix satisfies:
 - $R_{z,0} = I$
 - $R_{z,\theta} R_{z,\phi} = R_{z,\theta+\phi}$
- The above two properties imply
$$R_{z,\theta}^{-1} = R_{z,-\theta}$$



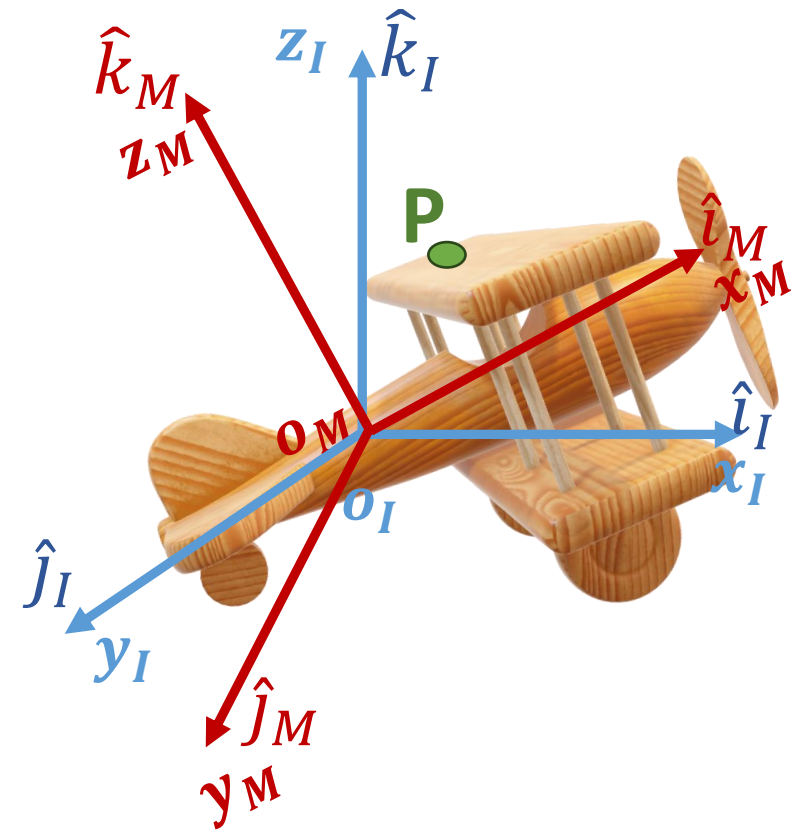
Example

- Similarly,

$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$R_{y,\theta} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

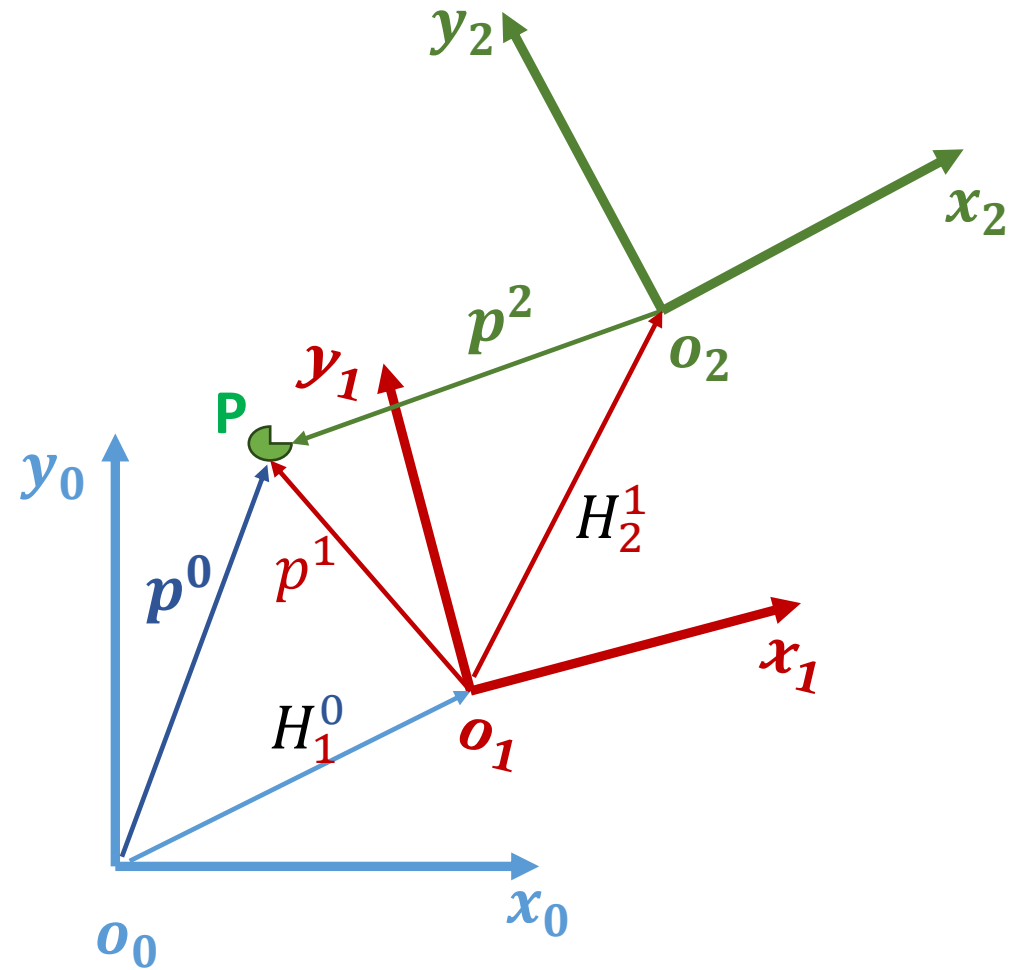
$$R_{z,\theta} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Composition of Rotations

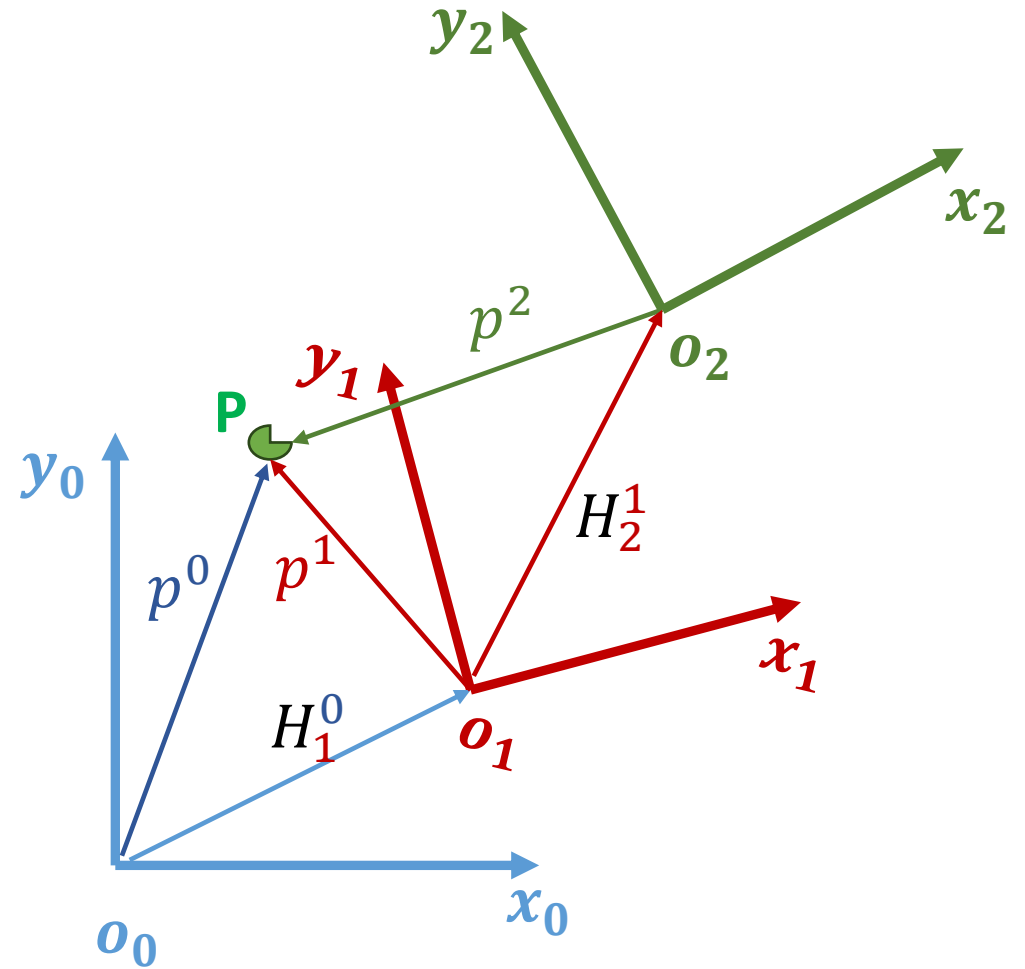
- Consider the setup in the figure with three frames:

- Frame 0: (o_0, x_0, y_0)
- Frame 1: (o_1, x_1, y_1)
- Frame 2: (o_2, x_2, y_2)



Composition of Rotations

- Consider the setup in the figure with three frames:
 - Frame 0: (o_0, x_0, y_0)
 - Frame 1: (o_1, x_1, y_1)
 - Frame 2: (o_2, x_2, y_2)
- A point P can be expressed with respect to each of the three frames.
 - Frame 0: p^0
 - Frame 1: p^1
 - Frame 2: p^2



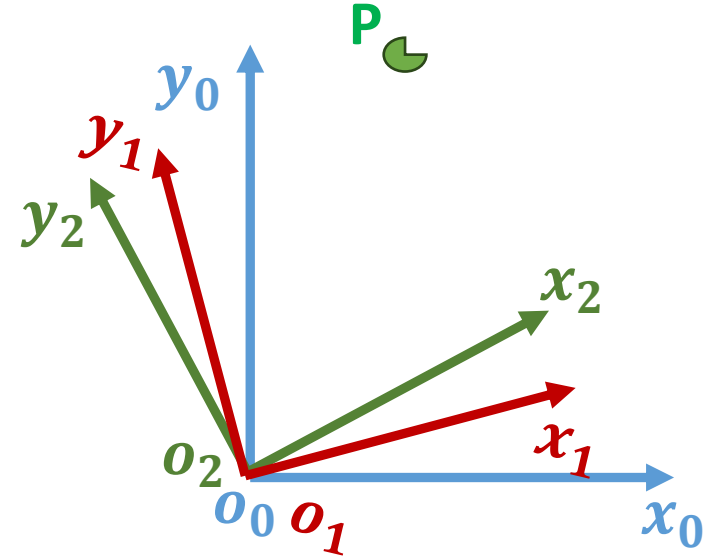
Composition of Rotations

- Consider the setup in the figure with three frames:

- Frame 0: (o_0, x_0, y_0)
- Frame 1: (o_1, x_1, y_1)
- Frame 2: (o_2, x_2, y_2)

- Let's start with the simple scenario in which the three frames have the same origin

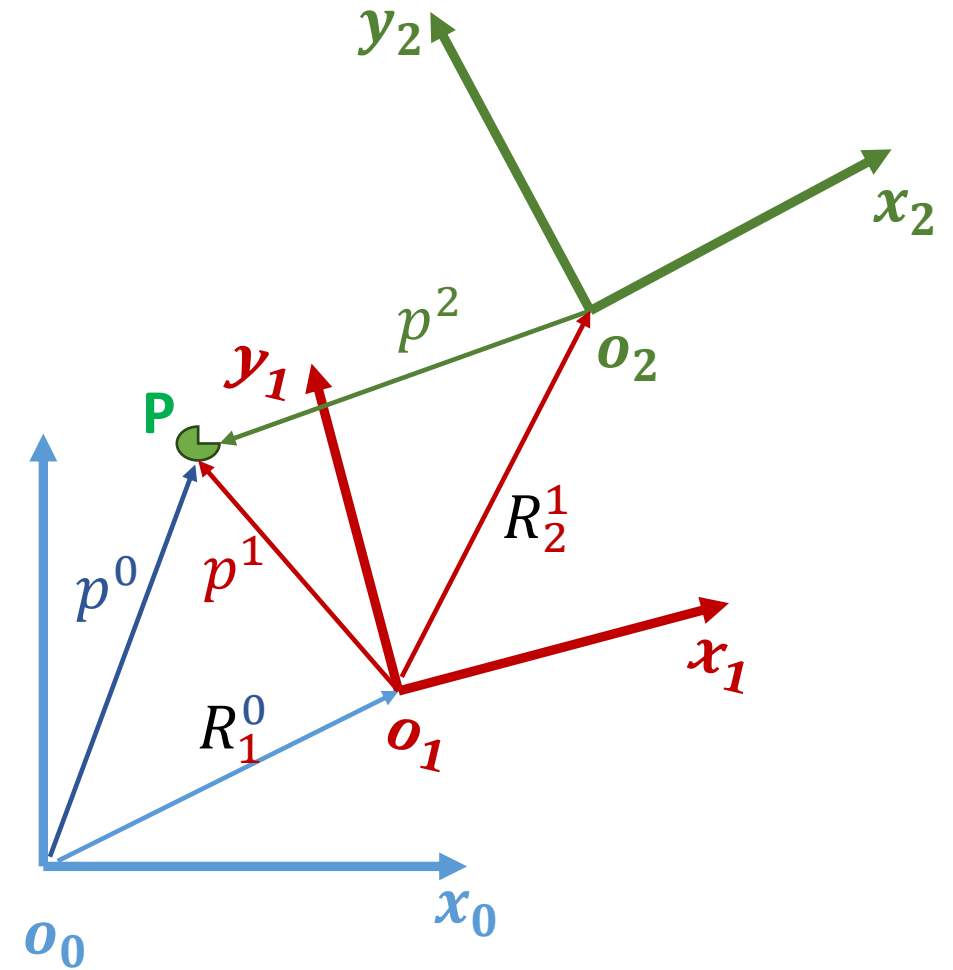
$$o_0 = o_1 = o_2$$



Composition of Rotations

- The three representations can be related through rotation matrices

- $p^0 = R_2^0 p^2$
- $p^0 = R_1^0 p^1$
- $p^1 = R_2^1 p^2$



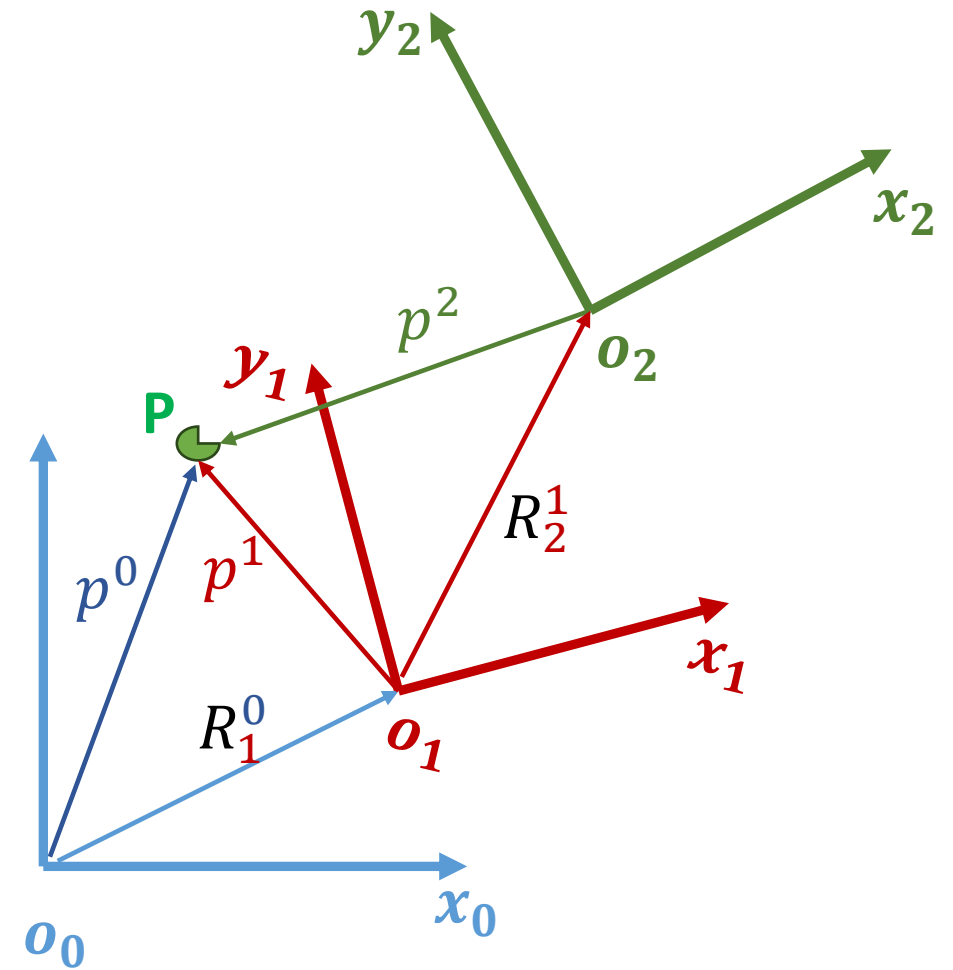
Composition of Rotations

- The three representations can be related through rotation matrices

- $p^0 = R_1^0 p^1$
- $p^0 = R_2^0 p^2$
- $p^1 = R_2^1 p^2$

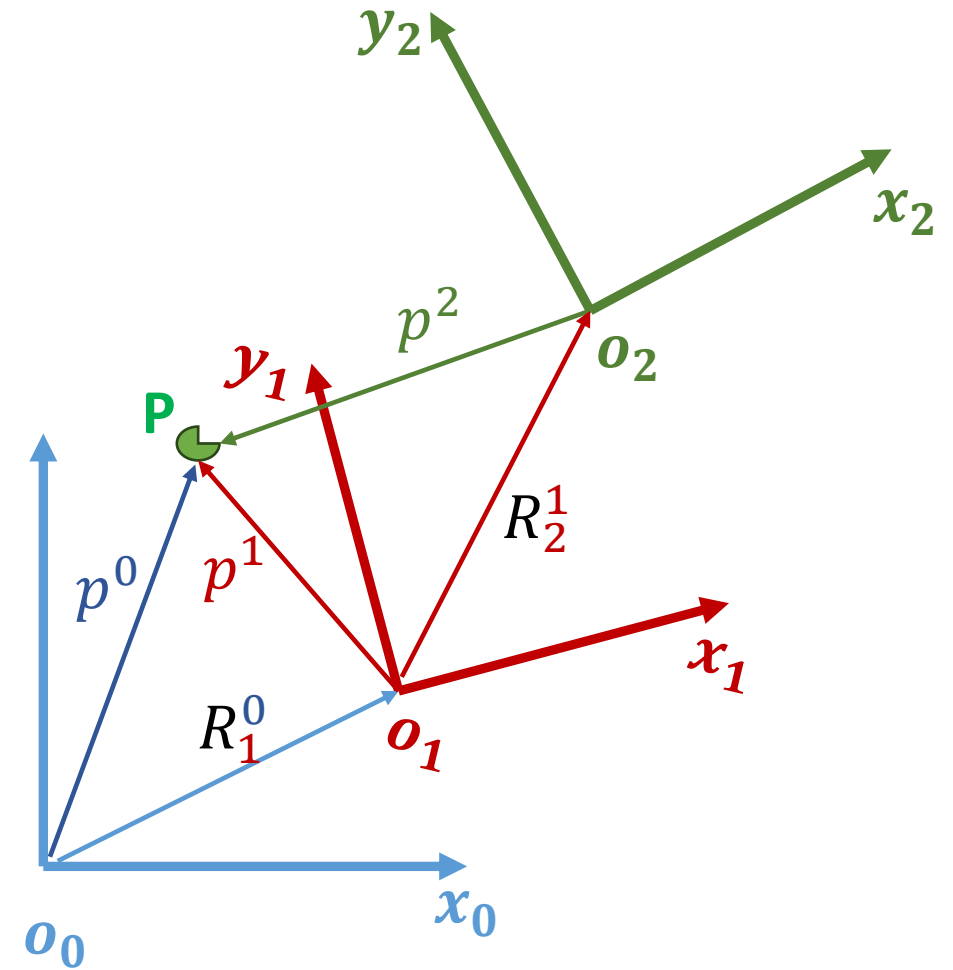
- From the above relations, we can relate p^0 to p^2 as follows:

$$p^0 = R_1^0 R_2^1 p^2$$



Composition of Rotations

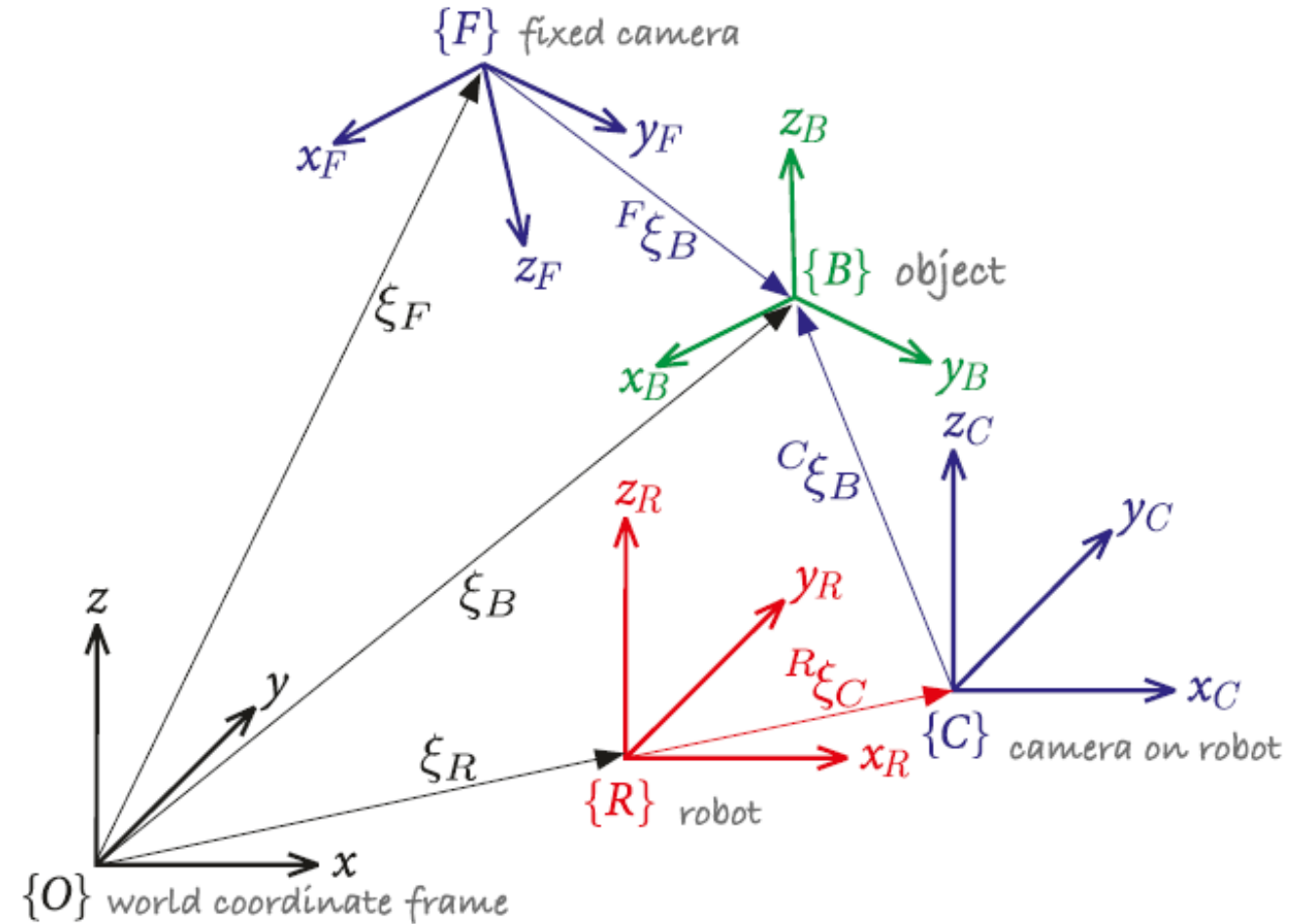
- The three representations can be related through rotation matrices
 - $p^0 = R_2^0 p^2$
 - $p^0 = R_1^0 p^1$
 - $p^1 = R_2^1 p^2$
- From the above relations, we can relate p^0 to p^2 as follows:
$$p^0 = R_1^0 R_2^1 p^2$$
- Thus, $R_2^0 = R_1^0 R_2^1$ is the **compositional law** for rotational transformation.



Composition of Rotations

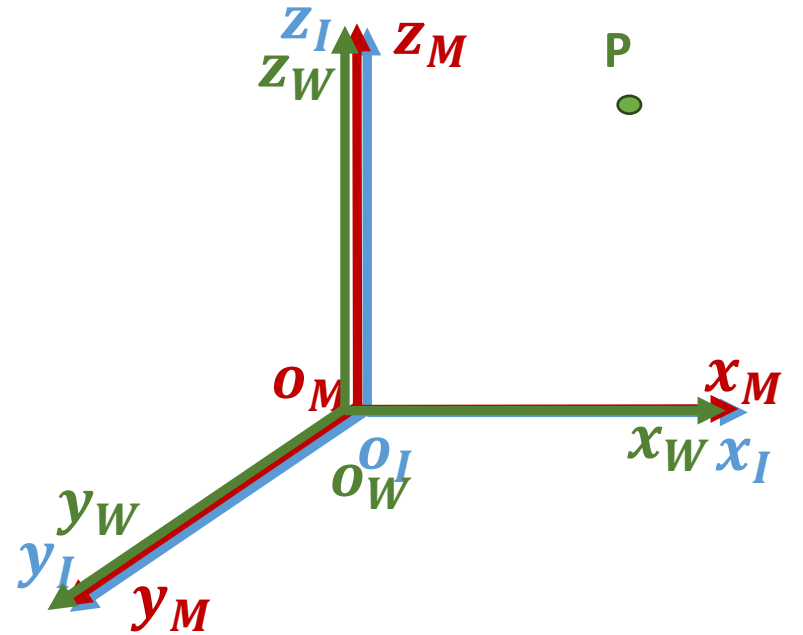
An example where coordinate transforms are practically useful:

- Suppose there is an **object** that is detected by a fixed camera, but not in the robot camera.
- The coordinates may need to be transformed to the **world frame** for global path planning, or to the **robot frame** for local decision making.



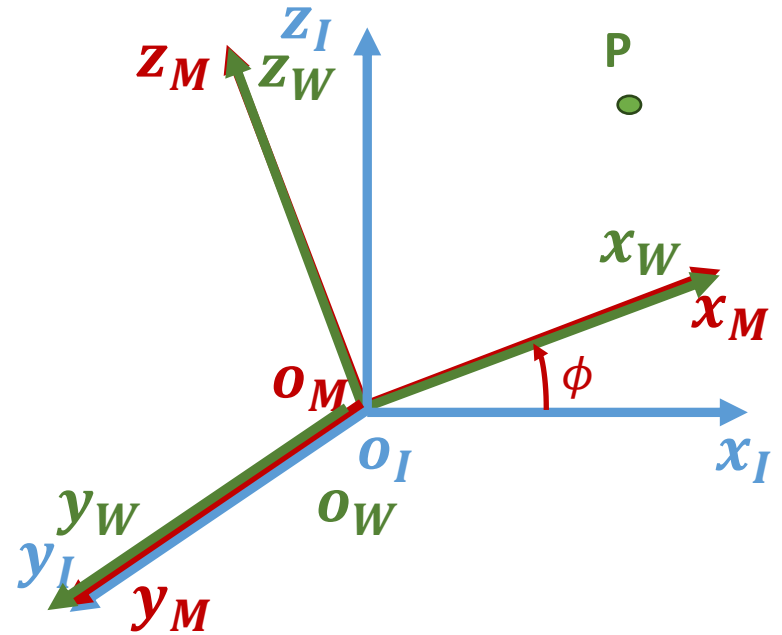
Rotation With Respect to Current Frame

- Consider the setup in the figure with three frames:
 - Frame I: (o_I, x_I, y_I, z_I)
 - Frame M: (o_M, x_M, y_M, z_M)
 - Frame W: (o_W, x_W, y_W, z_W)
- Suppose, initially, the three coordinate frames are **aligned**.



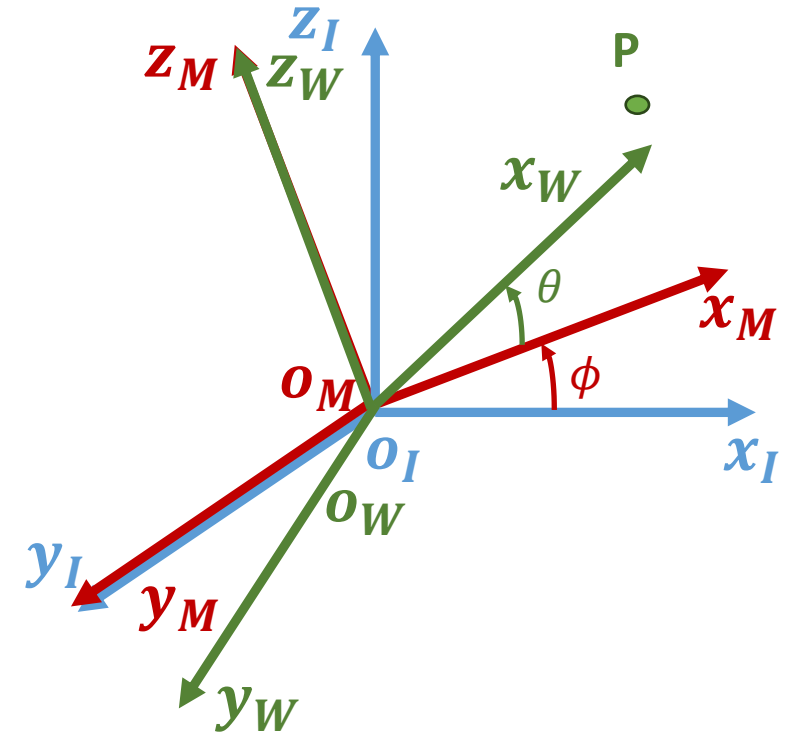
Rotation With Respect to Current Frame

- Suppose, initially, the three coordinate frames are aligned.
- We **first rotate** the **frame** (o_M, x_M, y_M, z_M) relative to (o_I, x_I, y_I, z_I) according to the transformation $R_1^0 = R_{y,\phi}$

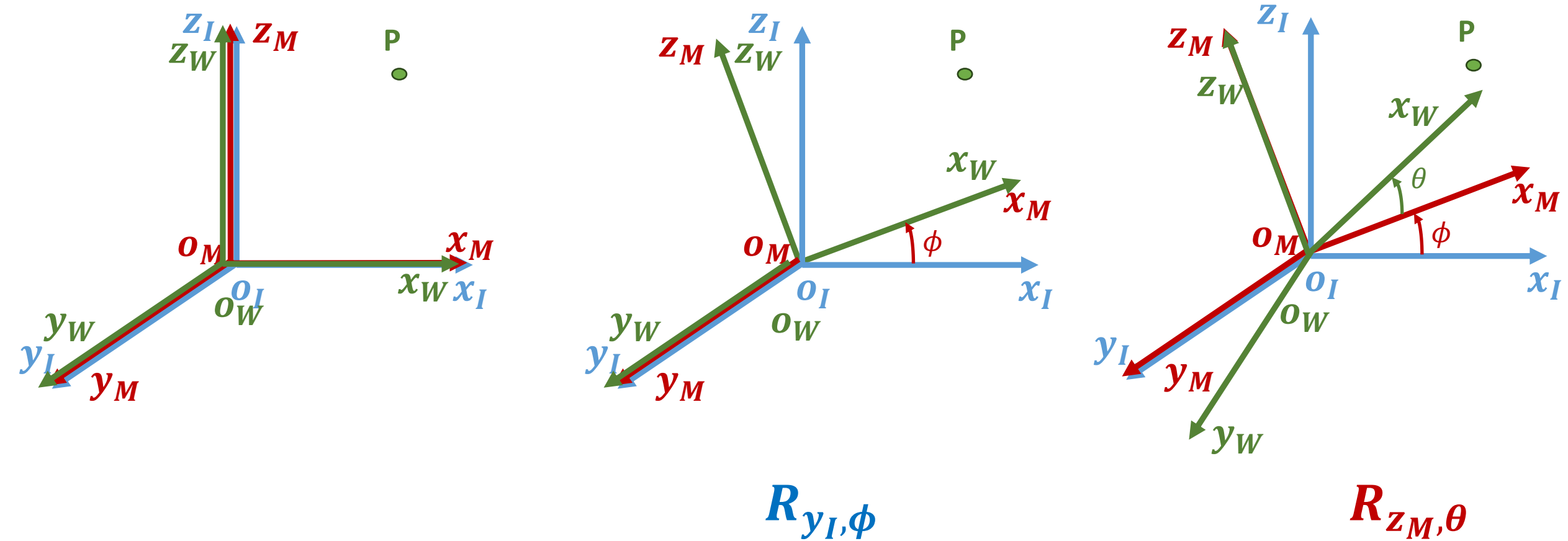


Rotation With Respect to Current Frame

- Suppose, initially, the three coordinate frames are aligned.
- We **first rotate** the **frame** (o_M, x_M, y_M, z_M) relative to (o_I, x_I, y_I, z_I) according to the **transformation** $R_1^0 = R_{y,\phi}$
- Then, with the frames (o_M, x_M, y_M, z_M) and (o_W, x_W, y_W, z_W) coincident, we rotate (o_W, x_W, y_W, z_W) relative to (o_M, x_M, y_M, z_M) according to **transformation** $R_2^1 = R_{z,\theta}$
- Since **Frame W** is rotated relative to **Frame M**, with which it was currently aligned, we say that **Frame W** is rotated relative to **current frame**.



Rotation With Respect to Current Frame



Example

- Rotation of ϕ degrees about current y-axis following by a rotation of θ degrees about current z-axis.

Example

- Rotation of ϕ degrees about current y-axis following by a rotation of θ degrees about current z-axis.
- The composite rotation matrix R is given by

$$\bullet R = R_{y\phi} R_{z\theta} = \begin{bmatrix} c_\phi & 0 & s_\phi \\ 0 & 1 & 0 \\ -s_\phi & 0 & c_\phi \end{bmatrix} \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\bullet R = \begin{bmatrix} c_\phi c_\theta & -c_\phi s_\theta & s_\phi \\ s_\theta & c_\theta & 0 \\ -s_\phi c_\theta & s_\phi s_\theta & c_\phi \end{bmatrix}$$

Example

- Rotation of ϕ degrees about current y-axis following by a rotation of θ degrees about current z-axis.

- The composite rotation matrix R is given by

$$\bullet R = R_{y\phi} R_{z\theta} = \begin{bmatrix} c_\phi & 0 & s_\phi \\ 0 & 1 & 0 \\ -s_\phi & 0 & c_\phi \end{bmatrix} \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\bullet R = \begin{bmatrix} c_\phi c_\theta & -c_\phi s_\theta & s_\phi \\ s_\theta & c_\theta & 0 \\ -s_\phi c_\theta & s_\phi s_\theta & c_\phi \end{bmatrix}$$

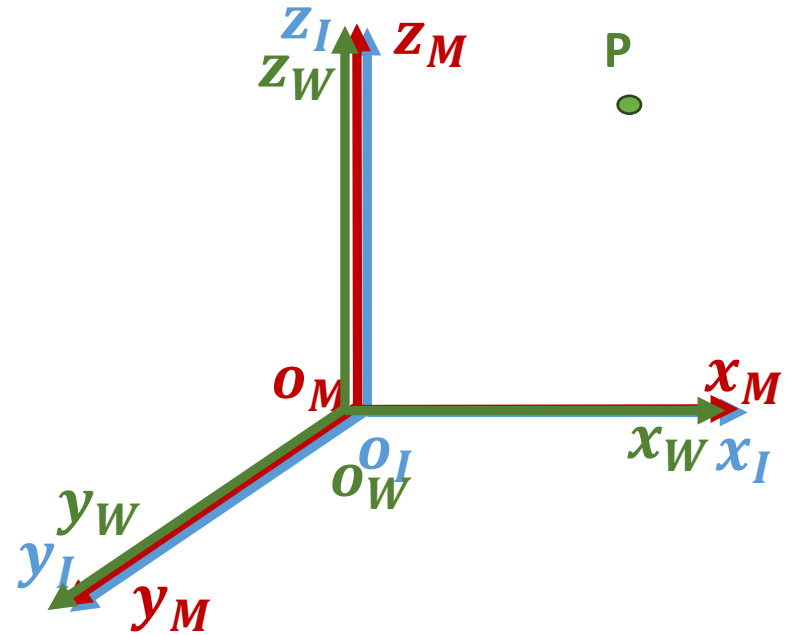
$$\bullet R' = R_{z\theta} R_{y\phi} = \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\phi & 0 & s_\phi \\ 0 & 1 & 0 \\ -s_\phi & 0 & c_\phi \end{bmatrix} = \begin{bmatrix} c_\theta c_\phi & -s_\theta & c_\theta s_\phi \\ s_\theta c_\phi & c_\theta & s_\theta s_\phi \\ -s_\phi & 0 & c_\phi \end{bmatrix}$$

Example

- Rotation of ϕ degrees about current y-axis following by a rotation of θ degrees about current z-axis.
- The composite rotation matrix R is given by
- $R = R_{y\phi} R_{z\theta} = \begin{bmatrix} c_\phi & 0 & s_\phi \\ 0 & 1 & 0 \\ -s_\phi & 0 & c_\phi \end{bmatrix} \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $R = \begin{bmatrix} c_\phi c_\theta & -c_\phi s_\theta & s_\phi \\ s_\theta & c_\theta & 0 \\ -s_\phi c_\theta & s_\phi s_\theta & c_\phi \end{bmatrix}$
- $R' = R_{z\theta} R_{y\phi} = \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\phi & 0 & s_\phi \\ 0 & 1 & 0 \\ -s_\phi & 0 & c_\phi \end{bmatrix} = \begin{bmatrix} c_\theta c_\phi & -s_\theta & c_\theta s_\phi \\ s_\theta c_\phi & c_\theta & s_\theta s_\phi \\ -s_\phi & 0 & c_\phi \end{bmatrix}$
- We can see that $R \neq R'$.
- Rotations are not commutative. Order of rotations is important.

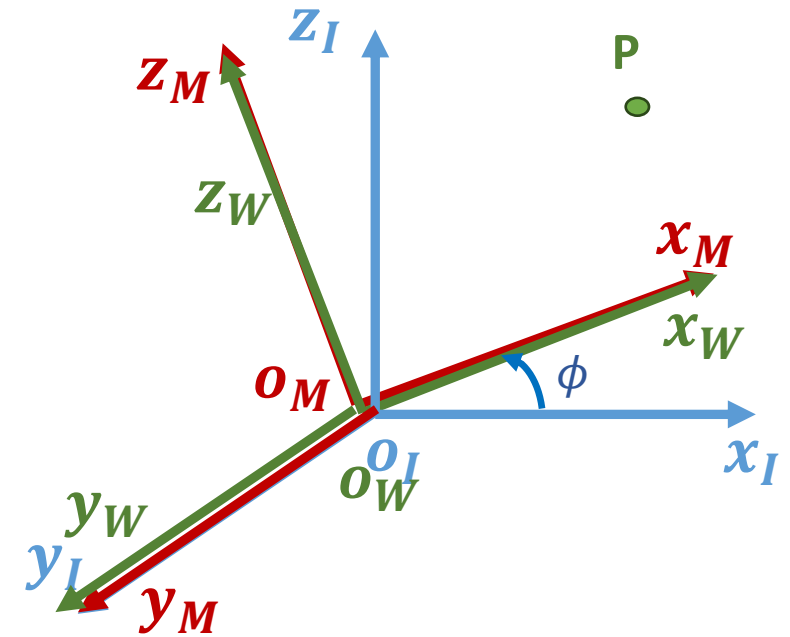
Rotation with respect to fixed frame

- Suppose we need a rotation of ϕ degrees about Y_I axis followed by a rotation of θ degrees about Z_I axis.
- As before, we assume that the three frames are initially aligned and their origins are the same.



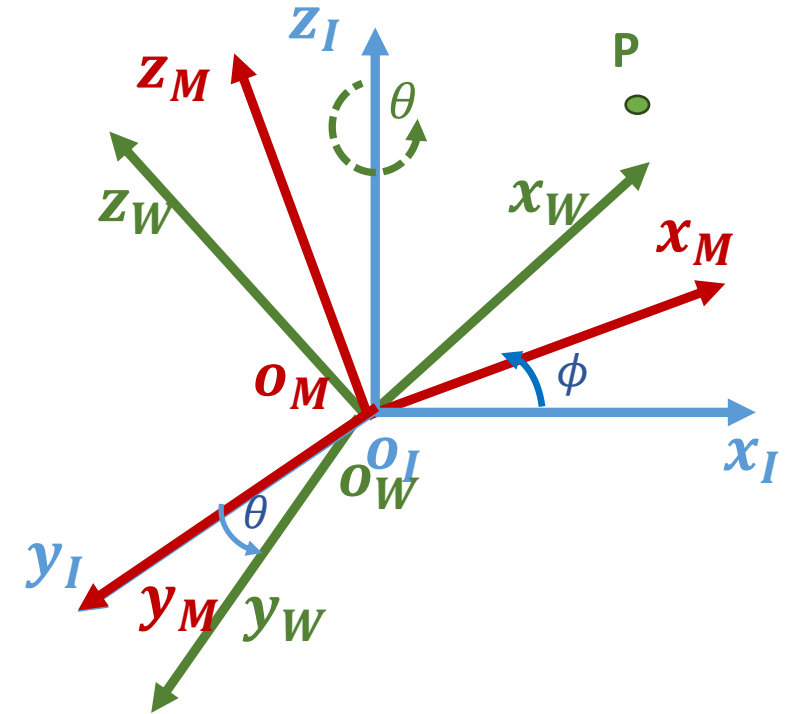
Rotation with respect to fixed frame

- In the first step, **Frame M**(o_M, x_M, y_M) is rotated along y_I axis by an angle ϕ .



Rotation with respect to fixed frame

- In the first step, **Frame M** (o_M, x_M, y_M) is rotated along y_I axis by an angle ϕ .
- In the next step, **Frame W** is rotated by an angle θ but along z_I axis.

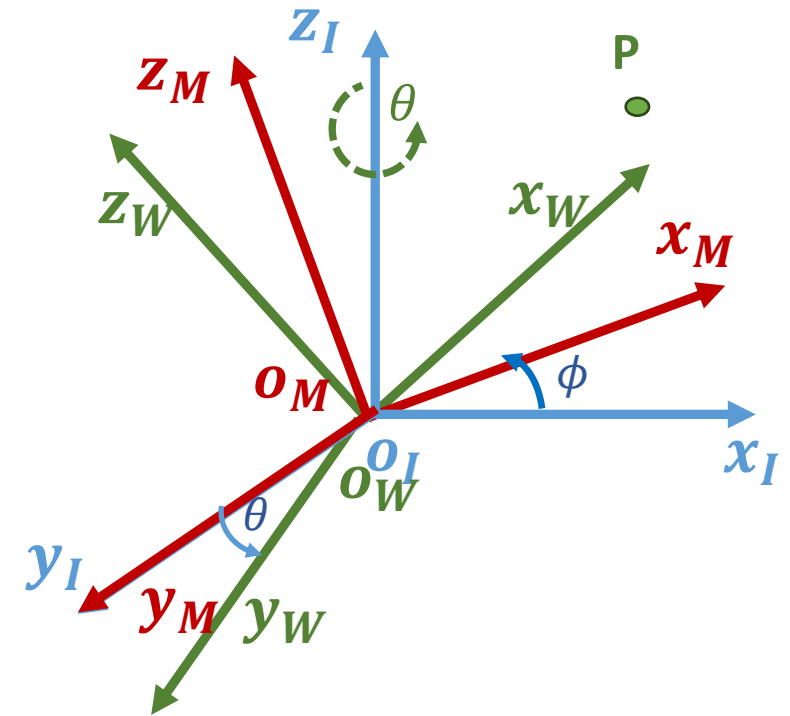


Rotation with respect to fixed frame

- Let p^I , p^M , and p^W correspond to the representation of these points with respect to the three frames.
- Then,

$$p^I = R_{y,\phi} p^M$$

where $R_{y,\phi} = R_M^I$ is basic rotation about y axis.



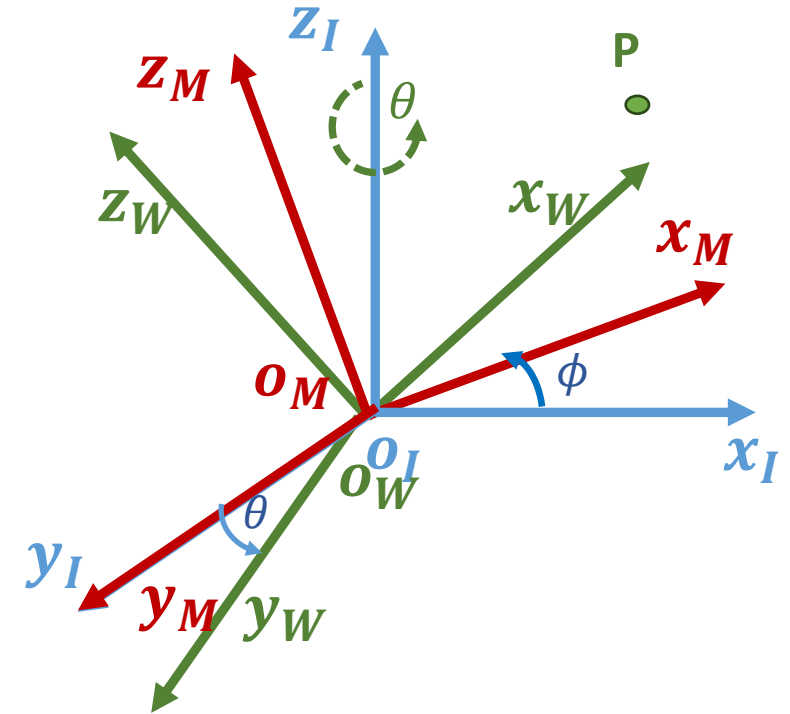
Rotation with respect to fixed frame

- Let p^I , p^M , and p^W correspond to the representation of point P with respect to the three frames.
- Then,

$$p^I = R_{y,\phi} p^M$$

where $R_{y,\phi} = R_M^I$ is basic rotation about y axis.

- What about the relation between p^M and p^W
 $p^M = R_{z,\theta} p^W$???



Rotation with respect to fixed frame

- Let p^I , p^M , and p^W correspond to the representation of point P with respect to the three frames.
- Then,

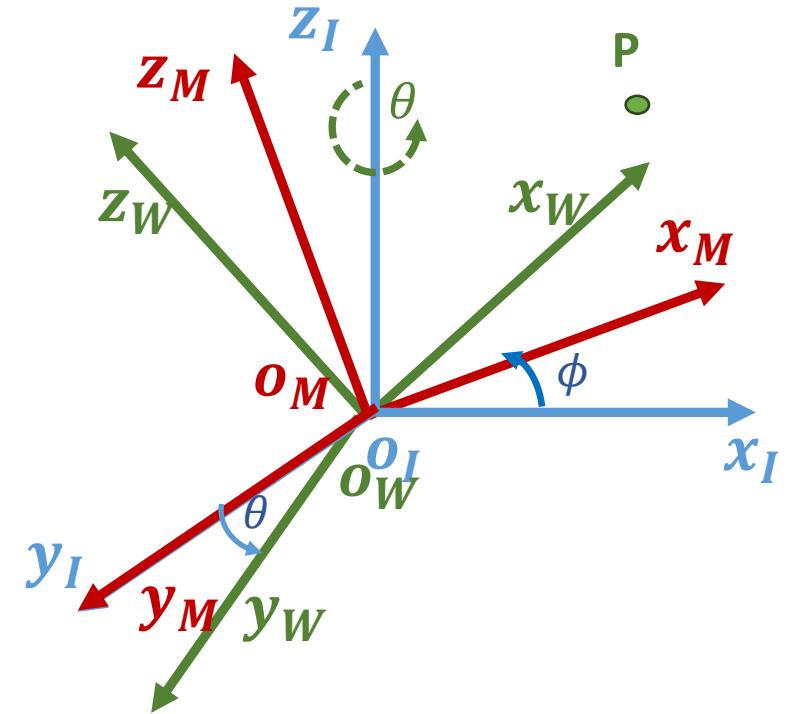
$$p^I = R_{y,\phi} p^M$$

where $R_{y,\phi} = R_M^I$ is basic rotation about y axis.

- Since rotation is with respect to the inertial frame, we cannot write

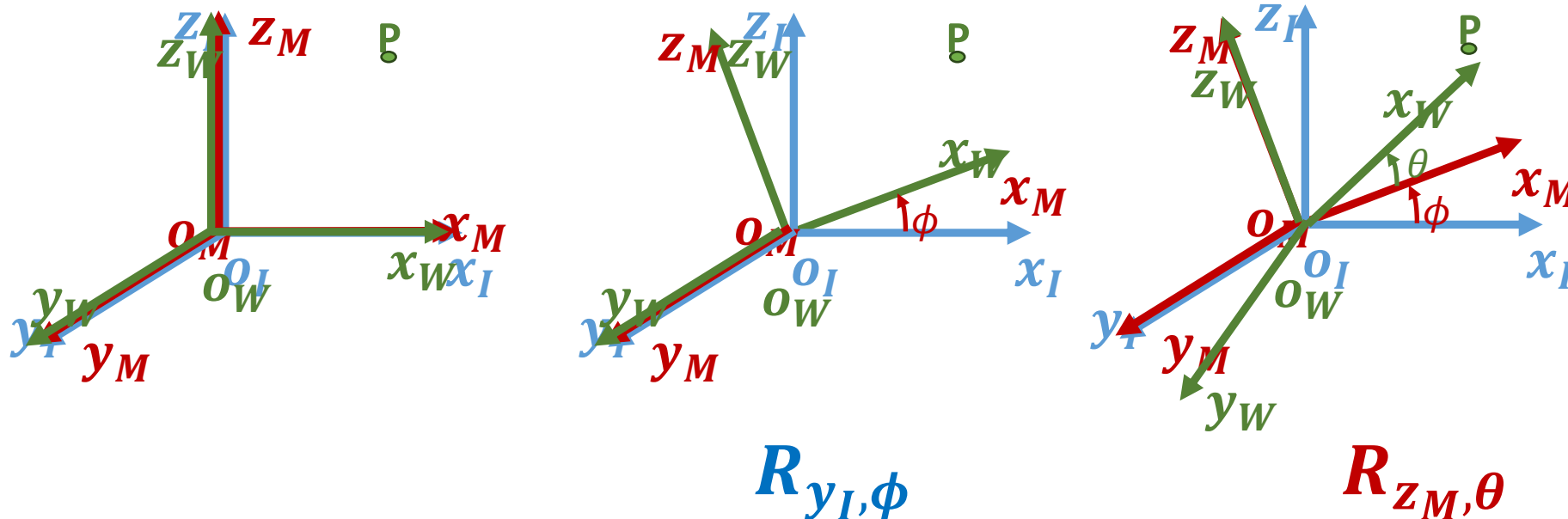
$$\boxed{p^M \neq R_{z,\theta} p^W}$$

because it represents rotation about the current z-axis.



Similarity Transformation

- A coordinate frame is defined by a **set of basis vectors**.
- A **Rotation Matrix**, as a **coordinate frame transformation**, can be viewed as a **change of basis from one frame to another**.
- The matrix representation of a **general linear transformation** is transformed from one frame to another using a **Similarity Transformation**.



Similarity Transformation

- Two matrices A and B are similar if and only if there exists an invertible matrix S such as

$$B = S^{-1} A S$$

- Two matrices are similar if and only if they represent the same linear map under two possibly different bases, with S being the change-of-basis matrix.
- If two matrices are similar, their eigenvalues are the same.

Similarity Transformation

- Suppose, matrix A represents a given linear transformation in the base frame (o_I, x_I, y_I, z_I) .
- Let matrix B represents the same transformation in mobile frame (o_M, x_M, y_M, z_M) .
- Then, A and B are related as

$$B = (R_M^I)^{-1} A R_M^I$$

- Here R_M^I is the coordinate transformation from the base frame (o_I, x_I, y_I, z_I) to mobile frame (o_M, x_M, y_M, z_M) .

Similarity Transformation

Example:

- Suppose frames (o_I, x_I, y_I, z_I) and frame (o_M, x_M, y_M, z_M) are related by the rotation matrix

$$R_M^I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = R_{y, \frac{\pi}{2}}$$

- If $A = R_{z, \theta}$ relative to the frame (o_I, x_I, y_I, z_I) , then, relative to frame (o_M, x_M, y_M, z_M) , we have

$$B = (R_M^I)^{-1} A R_M^I = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} c_\theta & s_\theta & 0 \\ -s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$
$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\theta & s_\theta \\ 0 & -s_\theta & c_\theta \end{bmatrix}$$

- Thus, B is a rotation about z_I axis but expressed relative to the frame (o_M, x_M, y_M, z_M) .

Rotation with respect to fixed frame

- Let p^I , p^M , and p^W correspond to the representation of point P with respect to the three frames.
- Then,

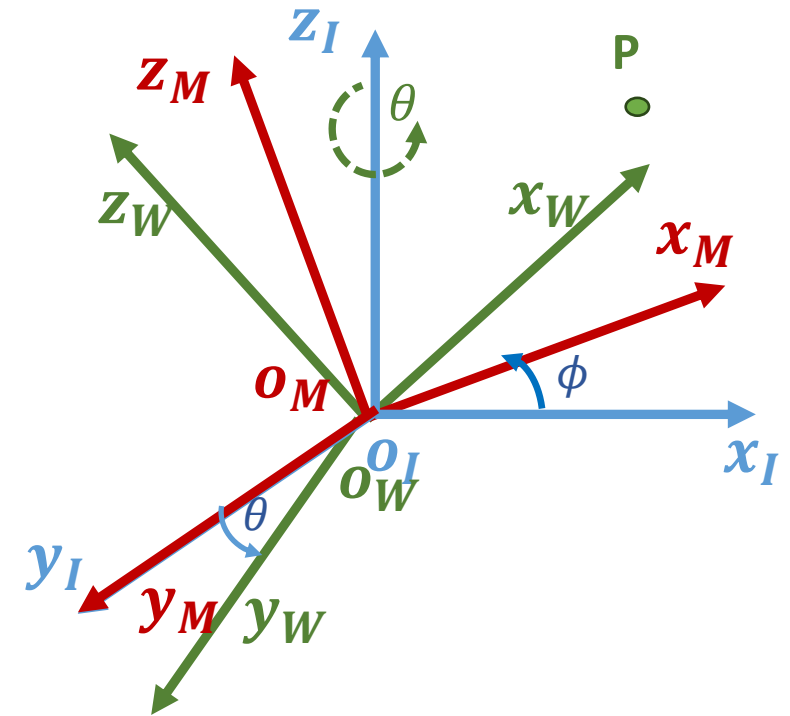
$$p^I = R_{y,\phi} p^M$$

where $R_{y,\phi} = R_M^I$ is basic rotation about y axis.

- Since rotation is with respect to the inertial frame, we cannot write

$$\boxed{p^M \neq R_{z,\theta} p^W}$$

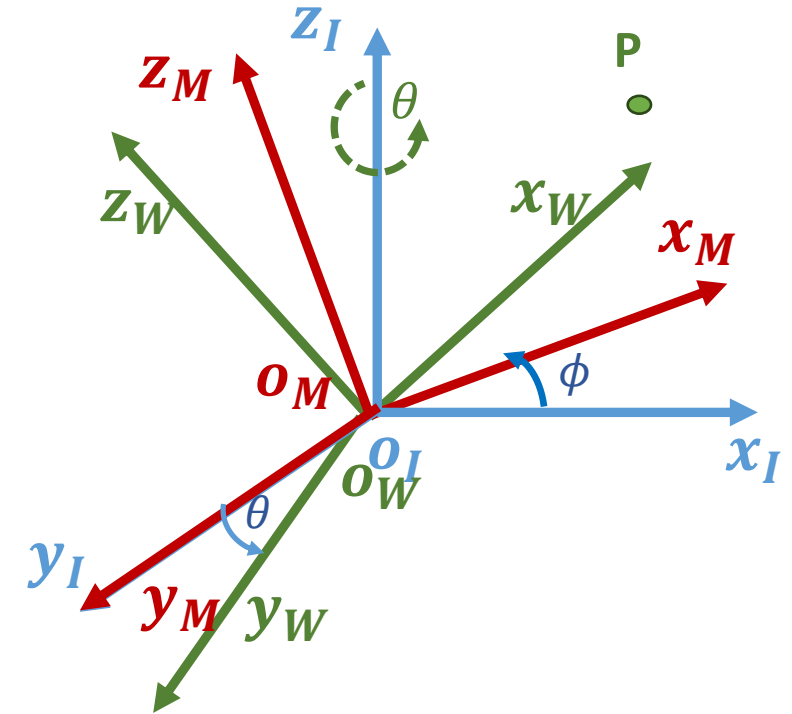
because it represents rotation about the current z-axis.



Rotation with respect to fixed frame

- Rotation of frame (o_W, x_W, y_W, z_W) is defined with respect to the inertial frame (o_I, x_I, y_I, z_I) .
- Our objective is to define this rotation, $R_{z,\theta}$, with respect to frame (o_M, x_M, y_M, z_M) .
- We know the relation between the base on mobile frames, which is $R_M^I = R_{y,\phi}$.
- The rotation $R_{z,\theta}$ which is defined with respect to (o_I, x_I, y_I, z_I) is

$$(R_M^I)^{-1} R_{z,\theta} R_M^I = R_{y,-\phi} R_{z,\theta} R_{y,\phi}$$
- Here, $R_{y,-\phi} R_{z,\theta} R_{y,\phi}$ is the rotation about z_I axis defined with respect to frame (o_M, x_M, y_M, z_M) .



Rotation with respect to fixed frame

- We know that $p^I = R_{y,\phi} p^M$. Then,

$$p^M = (R_{y,-\phi} R_{z,\theta} R_{y,\phi}) p^W$$

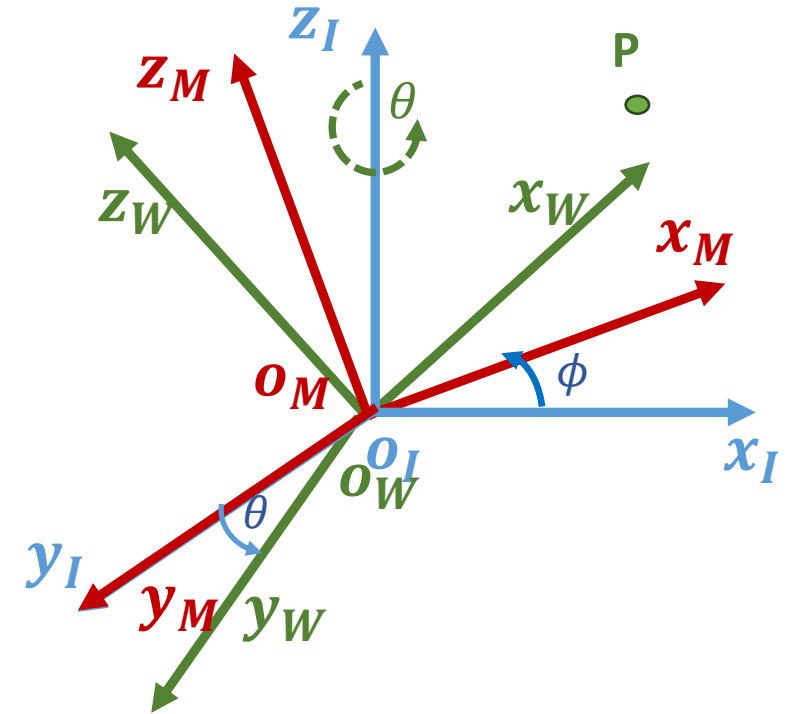
were, $R_{y,-\phi} R_{z,\theta} R_{y,\phi}$ is the rotation about z_I axis defined with respect to **frame** (o_M, x_M, y_M, z_M) .

- The composition rule for rotational transformation gives us

$$p^I = R_{y,\phi} p^M = R_{y,\phi} (R_{y,-\phi} R_{z,\theta} R_{y,\phi}) p^W$$

which results in

$$p^I = R_{z,\theta} R_{y,\phi} p^W$$



Current Frame Vs Fixed Frame

- Rotation of ϕ degrees about current y-axis following by a rotation of θ degrees about current z-axis.

$$p^I = R_M^I R_W^M p^W = R_{y_I, \phi} R_{z_M, \theta} p^W$$

- Rotation of ϕ degrees about inertial y_I -axis following by a rotation of θ degrees about inertial z_I -axis.

$$p^I = R_{z_I, \theta} R_{y_I, \phi} p^W$$

- The order of multiplication is **reversed** while composing rotations with respect to current frame vs fixed frame.

Parameterization of Rotations

- The nine elements r_{ij} in a general rotational transformation $R \in SO(3)$ are not independent quantities.
- Indeed, a rigid body possesses at most three rotational degrees of freedom, and thus at most three quantities required to specify its orientation.
- This can be easily seen by examining the constraints that govern the matrices in $SO(3)$

$$\sum_i r_{ij}^2 = 1 \text{ for all } j \in \{1,2,3\}$$
$$r_{1i}r_{1j} + r_{2i}r_{2j} + r_{3i}r_{3j} = 0, i \neq j$$

Parameterization of Rotations

- This can be easily seen by examining the **constraints** that govern the matrices in $SO(3)$

$$\sum_i r_{ij}^2 = 1 \text{ for all } j \in \{1,2,3\}$$

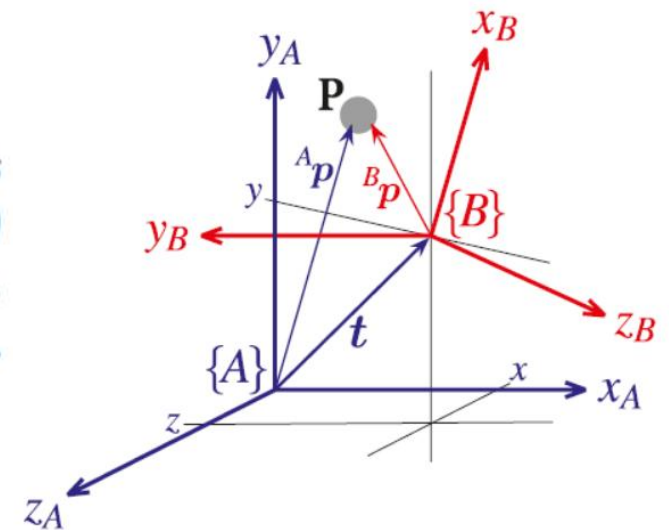
$$r_{1i}r_{1j} + r_{2i}r_{2j} + r_{3i}r_{3j} = 0, i \neq j$$

- The first set of constraints reflect the fact that **columns of a rotation matrix are unit vectors**.
- Columns of a **rotation matrix are mutually orthogonal**.
- Together, these constraints define **six independent equations** with **nine unknowns**, which implies that there are **three free variables**.

Euler Angles

- Imagine picking up frame A in our hand and rotating it until it looked just like frame B .

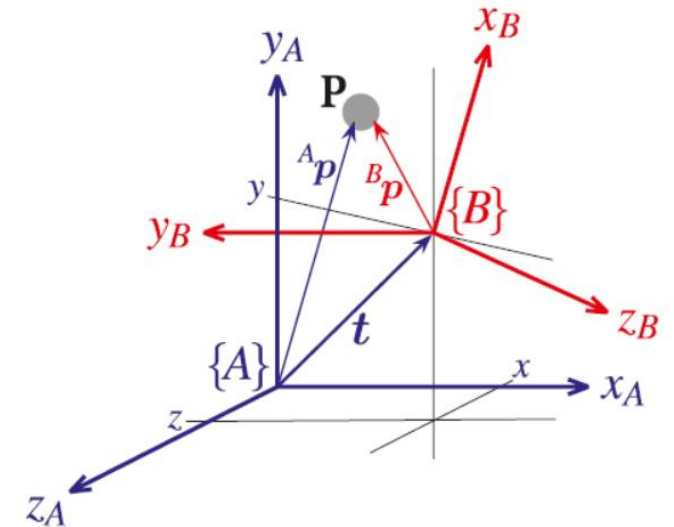
*Any two independent orthonormal coordinate frames can be related by a sequence of rotations (not more than **three**) about coordinate axes, where no two successive rotations may be about the same axis.*
Euler's rotation theorem (Kuipers 1999).



Euler Angles

*Any two independent orthonormal coordinate frames can be related by a sequence of rotations (not more than **three**) about coordinate axes, where no two successive rotations may be about the same axis.*
Euler's rotation theorem (Kuipers 1999).

- **Euler's rotation theorem** states that any rotation can be considered as a sequence of rotations about different coordinate axes.



*Any two independent orthonormal coordinate frames
can be related by a sequence of rotations (not more than three)
about coordinate axes, where no two successive rotations may be about the same axis.*
Euler's rotation theorem (Kuipers 1999).

Euler Angles

- There are **twelve different permutations** of rotations about **three axes** that meet Euler's criteria:

XYX, XYZ, XZX, XZY, YXY, YXZ, YZX, YZY, ZXY, ZXZ, ZYX, ZYZ

- **Why 12?**
- For the **first rotation** we have **3 axes** to choose from.
- For the **second rotation** we can choose an **axis not equal to the first**, so only **two choices**.
- For the **third rotation** we can choose an axis not equal to the second, **so again only two choices**.
- The number of choices is therefore **$3 \times 2 \times 2 = 12$**

Euler Angles

- Rotations about **three axes** that meet Euler's criteria:
 $XYX, XYZ, XZX, XZY, YXY, YXZ, YZX, YZY, ZXY, ZXZ, ZYX, ZYZ$
- We can divide these twelve permutations into **two groups of six**:
- **Eulerian (after Euler)** involves repetition, but not successively, of rotations about one particular axis: $XYX, XZX, YXY, YZY, ZXZ, \text{ or } ZYZ$.
- **Cardanian** (after Cardano, also called Tait-Bryan angles) involve rotations about all three axes: $XYZ, XZY, YZX, YXZ, ZXY, \text{ or } ZYX$.

Euler Angles

- In some references **all twelve sequences** are referred to as **Euler angles**, but here we will consider just the **Eulerian sequences** from above as Euler angles (**XYX, XZX, YXY, YZY, ZXZ, or ZYZ**).
- There are six different sequences to choose from, and the particular angle sequence is a convention within a particular technological field.
- In aerospace the convention for **Euler angles** is **ZYZ** where the corresponding rotation matrix is

$$R(\phi, \theta, \psi) = R_z(\phi)R_y(\theta)R_z(\psi)$$

Roll Pitch and Yaw

- The Cardanian angles (XYZ, XZY, YZX, YXZ, ZXY, or ZYX) are also known as roll, pitch and yaw angles.
- Confusingly there are two different versions in common use, the sequences XYZ and ZYX.
- Text books are not at all consistent on this matter.
- If there is any pattern to the inconsistency it is that the
 - Mobile robot community (drones, ground vehicles) uses ZYX
 - Robot manipulator community uses XYZ.

Euler Angle Vs Roll Pitch and Yaw

- Euler angles represent rotations along two axes.
- RPY represent rotations along all three axes.
- Euler angles represent rotations with respect to Fixed/Current frame.
- RPY represent rotations with respect to Fixed/Current Frame.

$$R(\text{roll}, \text{pitch}, \text{yaw}) = R_z(\text{yaw})R_y(\text{pitch})R_x(\text{roll})$$

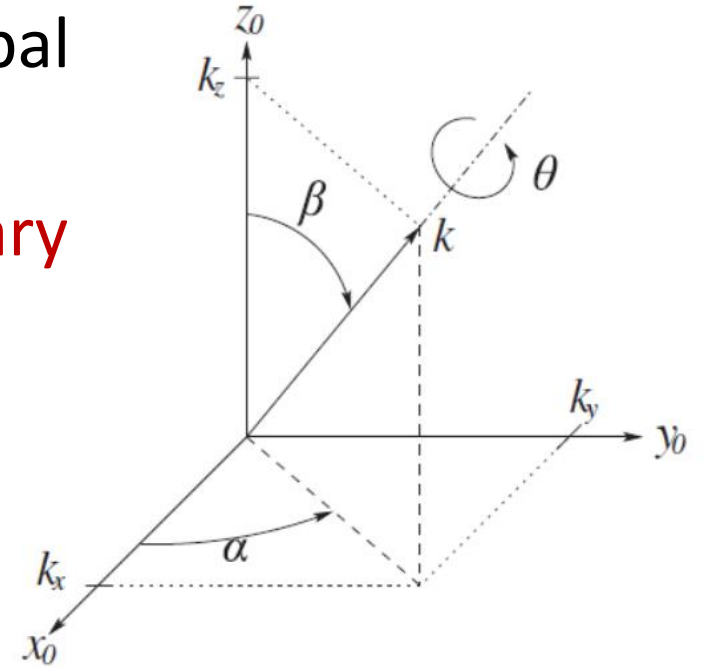
Rotation About an Arbitrary Axis

- Rotations are **not always performed** about the principal coordinate axes.
- We are often interested in a **rotation about an arbitrary axis** in space. This provides both a convenient way to describe rotations, and an alternative
- Let

$$\mathbf{k} = (k_x, k_y, k_z)$$

expressed in the frame (o_0, x_0, y_0, z_0) be a **unit vector defining an axis**.

- We wish to derive the **rotation matrix $R_{\mathbf{k}}$** representing a rotation of θ about this axis.

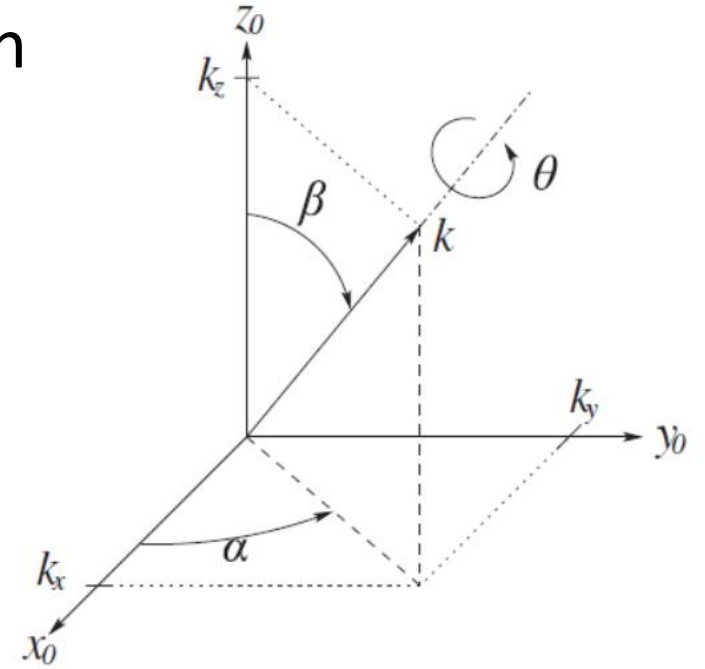


Rotation About an Arbitrary Axis

- There are several ways in which the **matrix $R_k(\theta)$** can be derived.
- One approach is to note that the rotational transformation

$$R = R_{z,\alpha} R_{y,\beta}$$

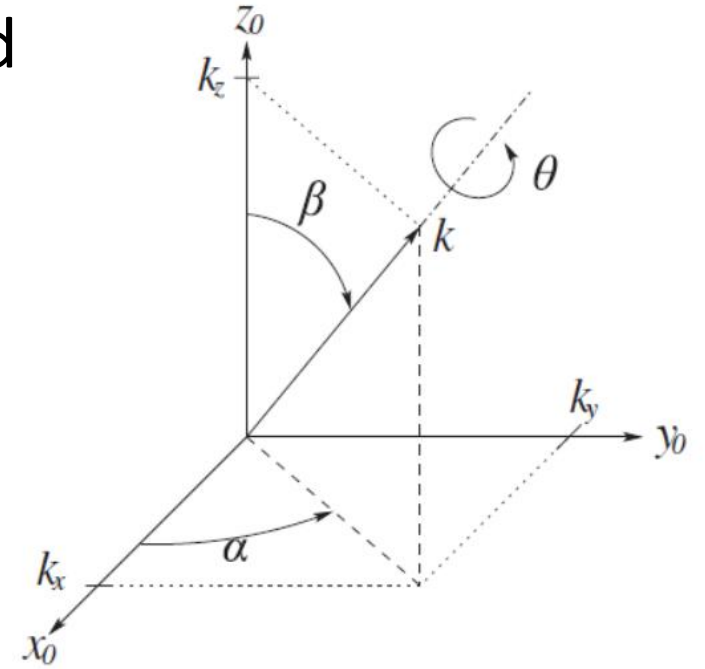
will bring the **world z – axis** into alignment with the **vector k** .



Rotation About an Arbitrary Axis

- Therefore, rotation about the axis k can be computed using a similarity transformation as

$$\begin{aligned} R_{k,\theta} &= R R_{z,\theta} R^{-1} \\ &= R_{z,\alpha} R_{y,\beta} R_{z,\theta} R_{y,-\beta} R_{z,-\alpha} \end{aligned}$$



Rotation About an Arbitrary Axis

- For computing the rotation matrix, we have

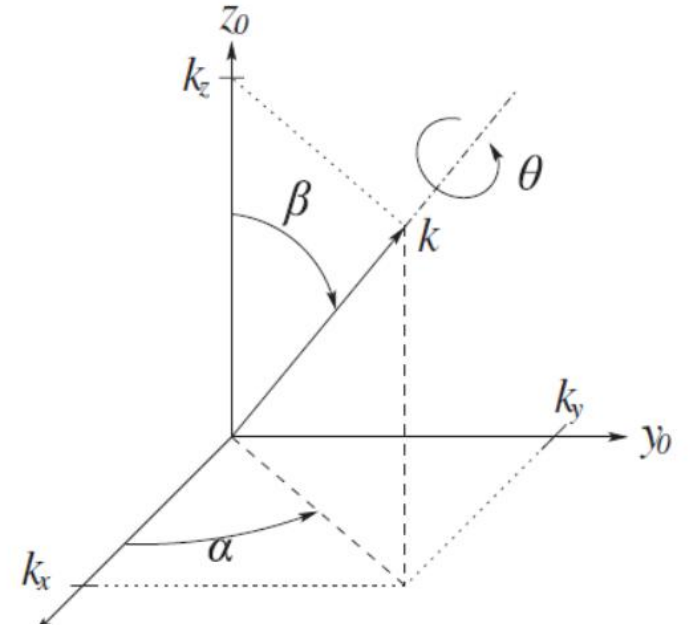
$$\sin \alpha = \frac{k_y}{\sqrt{k_x^2 + k_y^2}}, \quad \cos \alpha = \frac{k_x}{\sqrt{k_x^2 + k_y^2}}$$

$$\sin \beta = \frac{\sqrt{k_x^2 + k_y^2}}{\sqrt{k_x^2 + k_y^2 + k_z^2}}, \quad \cos \beta = \frac{k_z}{\sqrt{k_x^2 + k_y^2 + k_z^2}}$$

- After some length computations:

$$R_{k,\theta} = \begin{bmatrix} k_x^2 v_\theta + c_\theta & k_x k_y v_\theta - k_z s_\theta & k_x k_z v_\theta + k_y s_\theta \\ k_x k_y v_\theta + k_z s_\theta & k_y^2 v_\theta + c_\theta & k_y k_z v_\theta - k_x s_\theta \\ k_x k_z v_\theta - k_y s_\theta & k_y k_z v_\theta + k_x s_\theta & k_z^2 v_\theta + c_\theta \end{bmatrix}$$

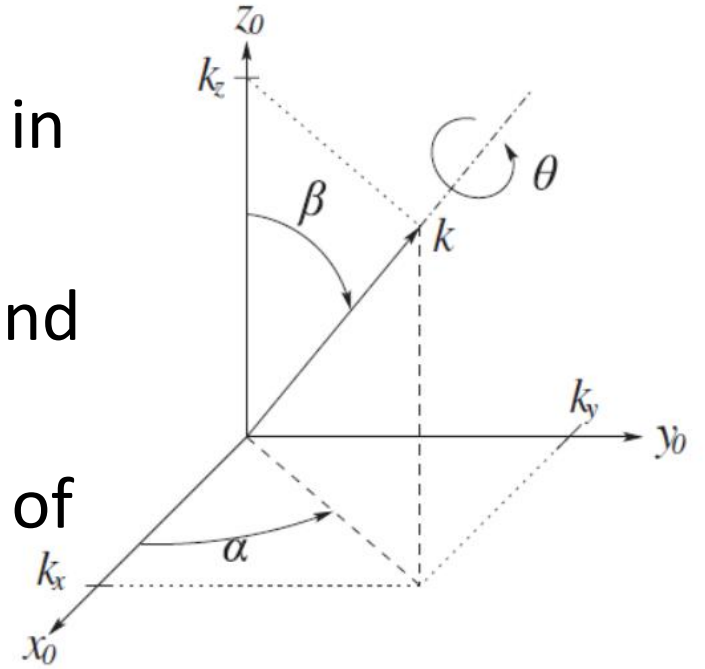
$$v_\theta = \text{vers } \theta = 1 - c_\theta$$



Rotation About an Arbitrary Axis

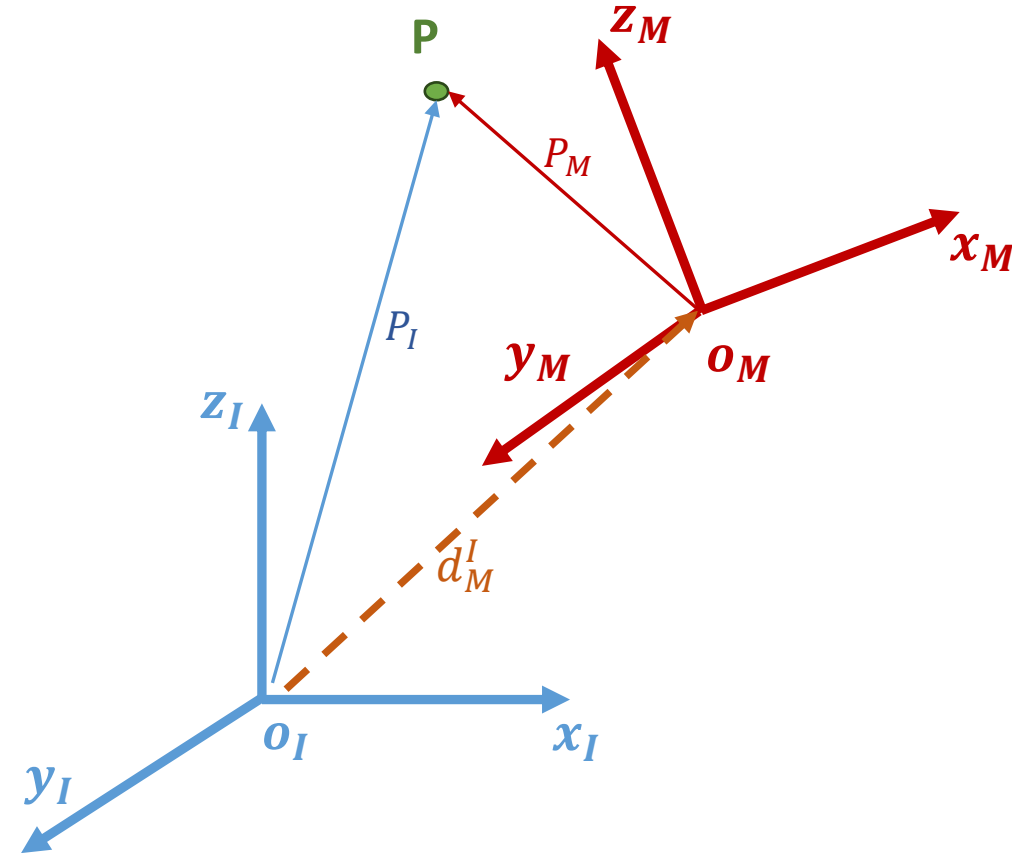
- In fact, **any rotation matrix** $R \in SO(3)$ can be represented by a **single rotation** about a suitable axis in space by a suitable angle $R = R_{k,\theta}$.
- Here k is a **unit vector** defining the axis of rotation, and θ is the **angle of rotation** about k .
- The pair (k, θ) is called the axis-angle representation of R .
- Given an arbitrary **rotation matrix** R with **components** r_{ij} , the equivalent angle θ and equivalent axis k are given by the expressions

$$\theta = \cos^{-1} \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right) \quad k = \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$



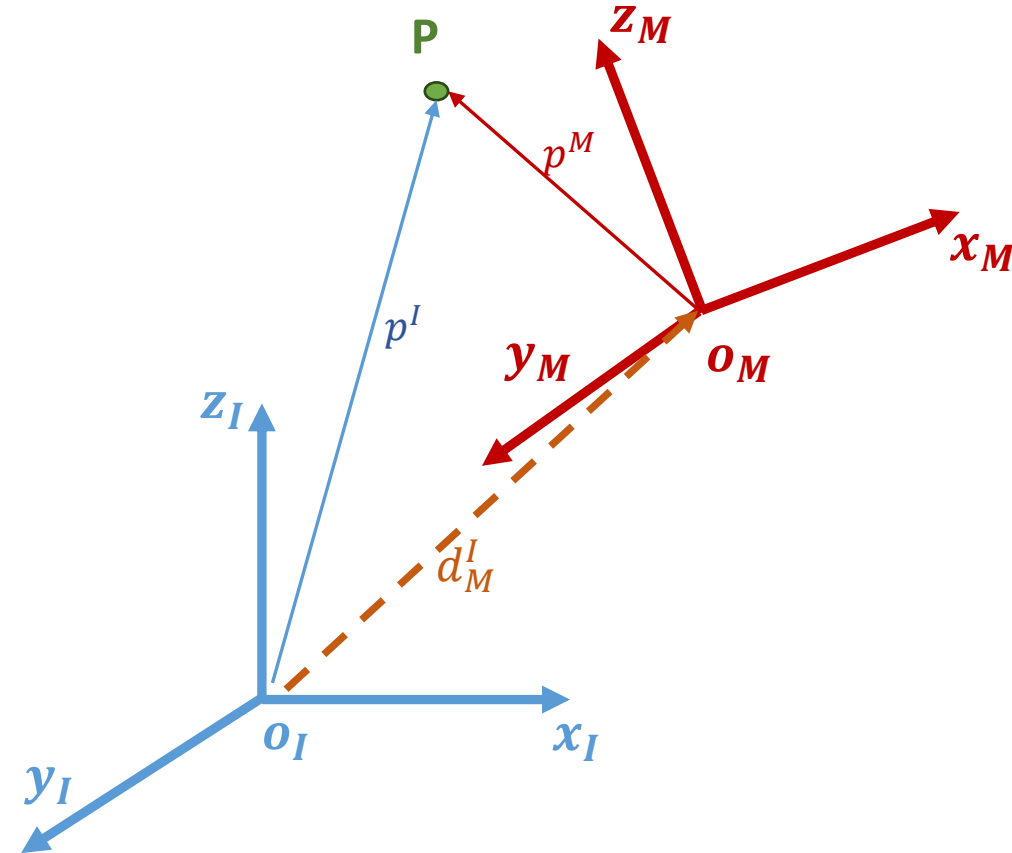
Rigid Body Motion

- A **rigid motion** is an **ordered pair** (d, R) where $d \in \mathbb{R}^3$ and $R \in SO(3)$.
- The group of all rigid motions is known as the **Special Euclidean Group** and is denoted by $SE(3)$, where $SE(3) = \mathbb{R}^3 \times SO(3)$.



Rigid Body Motion

- Let R_M^I be the rotation matrix that specifies the orientation of frame (o_M, x_M, y_M, z_M) with respect to (o_I, x_I, y_I, z_I) .
- Let d_M^I be the vector from the origin of frame (o_I, x_I, y_I, z_I) to the origin of frame (o_M, x_M, y_M, z_M) .
- Let p^I and p^M be the representations of a point P with respect to (o_I, x_I, y_I, z_I) and (o_M, x_M, y_M, z_M) , respectively.



Rigid Body Motion

- If the two frames were aligned, i.e., $R_M^I = I$, then

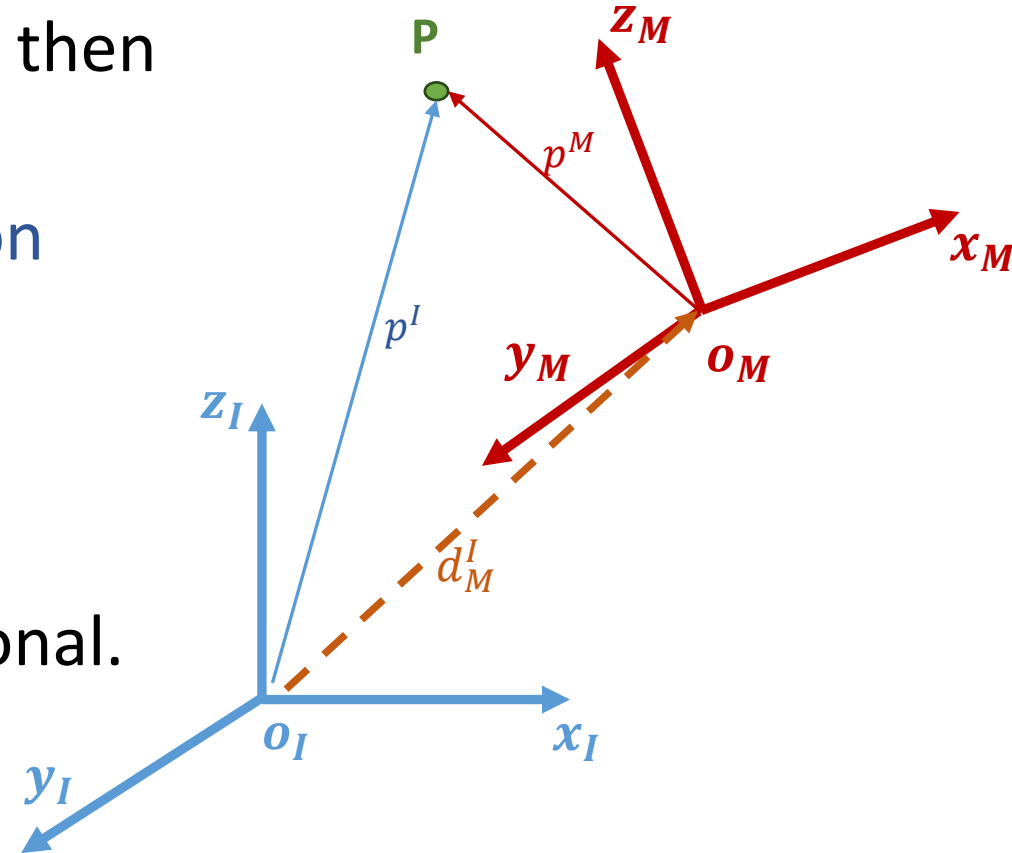
$$p^I = p^M + d_M^I$$

- In general, a **rigid motion** is a **pure translation** together with a **pure rotation**.

- A transformation

$$p^I = R_M^I p^M + d_M^I$$

is said to define a rigid motion if R_M^I is orthogonal.



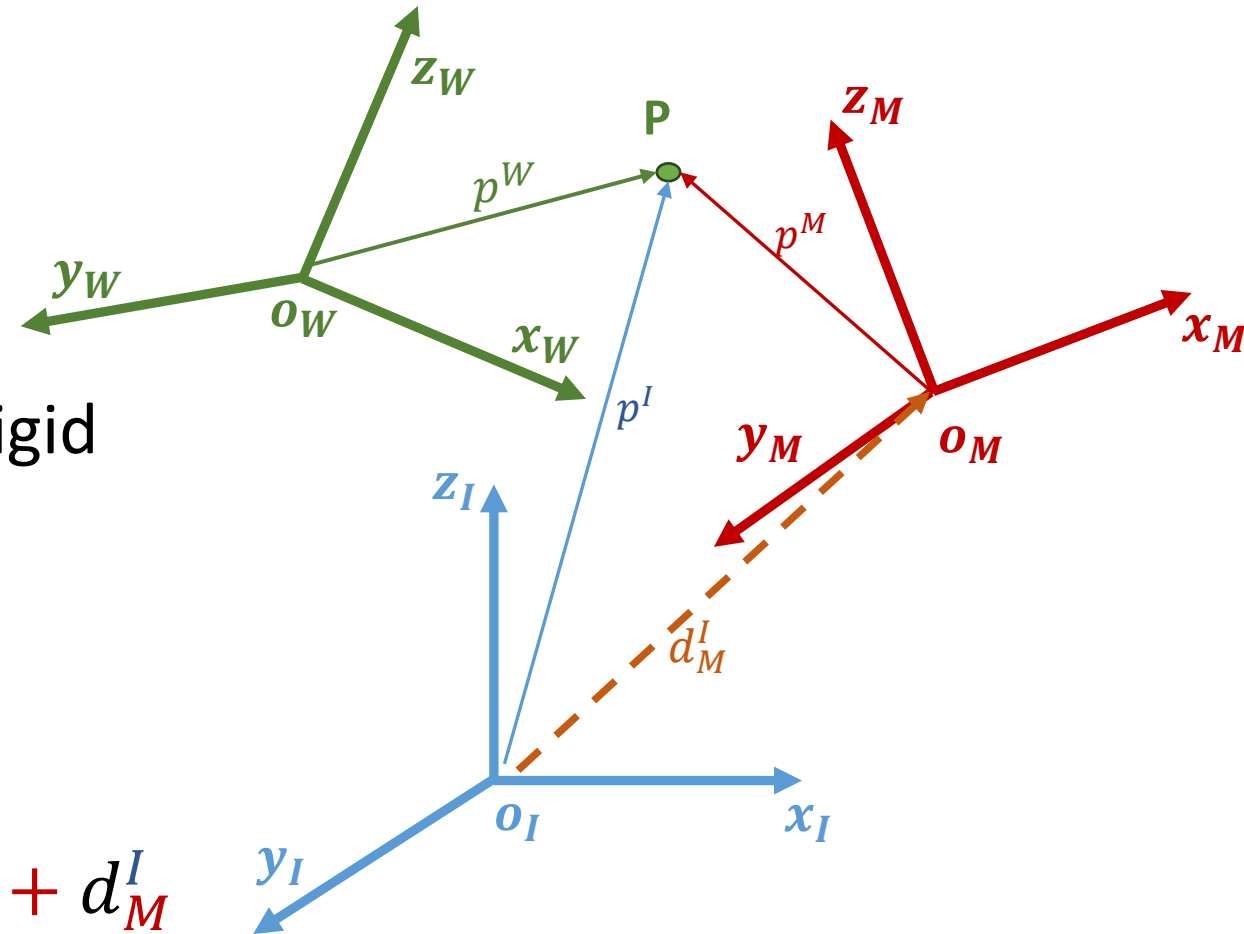
Rigid Body Motion

- If we have two rigid motions
 - $p^I = R_M^I p^M + d_M^I$
 - $p^M = R_W^M p^W + d_W^M$
- Then, the composition defines a third rigid motion, which is

$$p^I = R_M^I p^M + d_M^I$$

$$p^I = R_M^I (R_W^M p^W + d_W^M) + d_M^I$$

$$p^I = R_M^I R_W^M p^W + R_M^I d_W^M + d_M^I$$



Rigid Body Motion

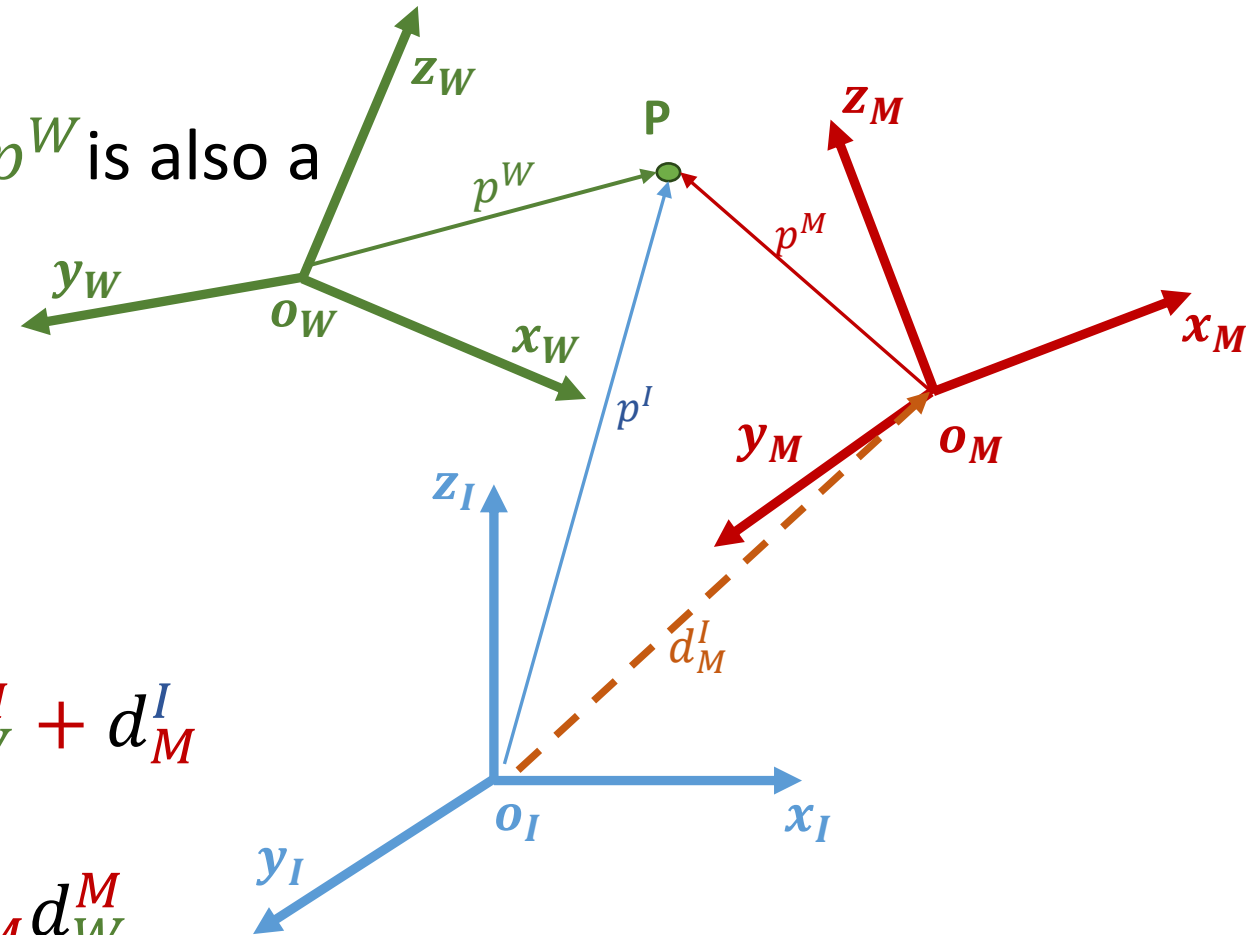
- Since the relationship between p^I and p^W is also a rigid motion, we can describe

$$p^I = R_W^I p^W + d_W^I$$

- Comparing with

$$p^I = R_M^I R_W^M p^W + R_M^I d_W^M + d_M^I$$

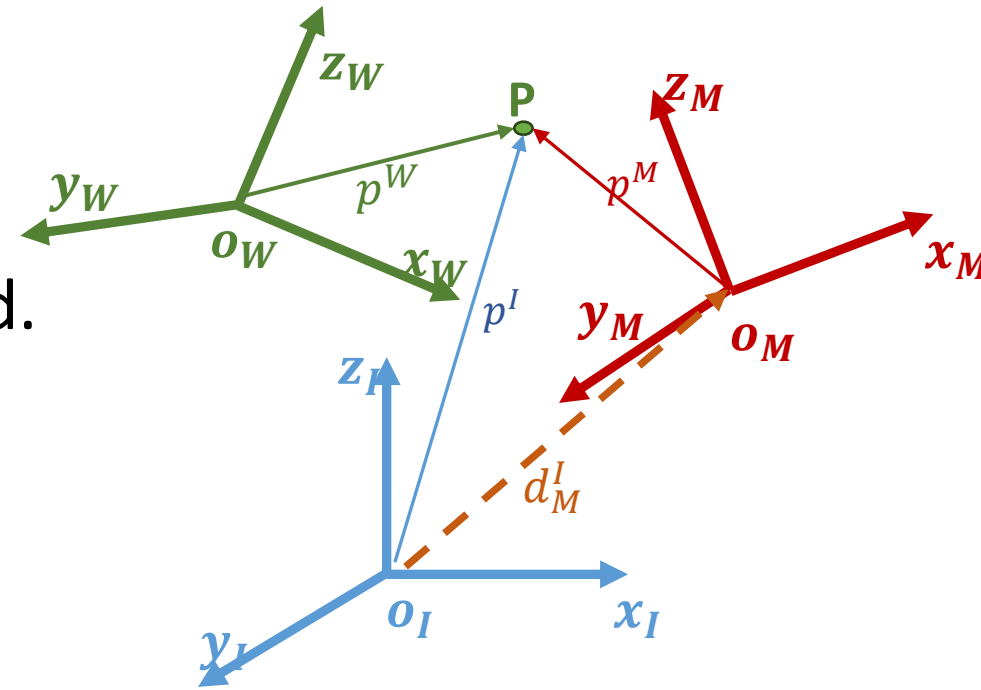
$$R_W^I = R_M^I R_W^M \text{ and } d_W^I = d_M^I + R_M^I d_W^M$$



Rigid Body Motion

$$R_{W}^I = R_M^I R_W^M \text{ and } d_{W}^I = d_M^I + R_M^I d_W^M$$

- Orientation transformations are simply multiplied.
- The translation vector must account for the translation d_M^I and translation from M to W frames expressed with respect to inertial frame.

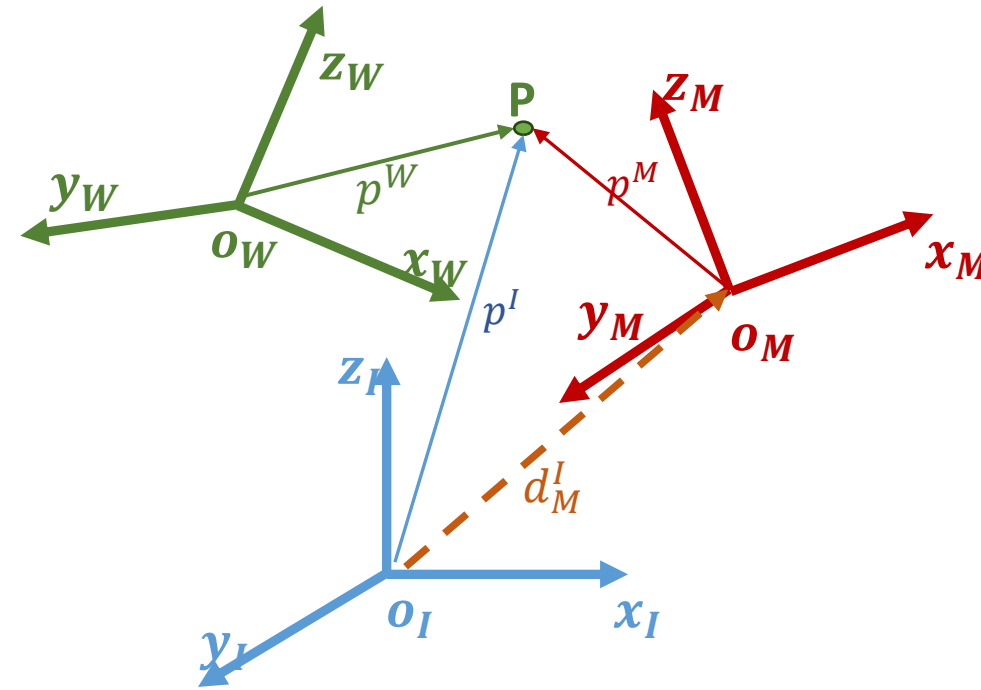


Rigid Body Motion

$$R_{W}^I = R_M^I R_W^M \text{ and } d_W^I = d_M^I + R_M^I d_W^M$$

- The composition of Homogeneous transformations can be expressed in matrix form as

$$\begin{bmatrix} R_M^I & d_M^I \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_W^M & d_W^M \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_M^I R_W^M & R_M^I d_W^M + d_M^I \\ 0 & 1 \end{bmatrix}$$



Rigid Body Motion

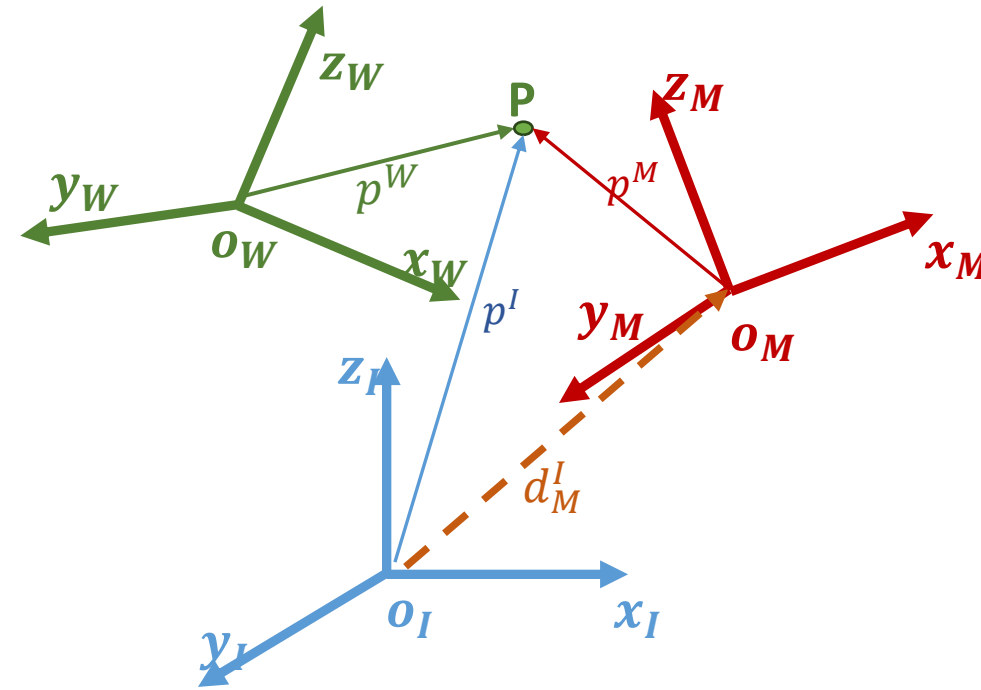
- In this matrix representation

$$\begin{bmatrix} R_M^I & d_M^I \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_W^M & d_W^M \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_M^I R_W^M & R_M^I d_W^M + d_M^I \\ 0 & 1 \end{bmatrix}$$

- $0 = [0 \ 0 \ 0]$ row vector.
- The set of matrices of the form

$$H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}, R \in SO(3)$$

represent rigid motions,



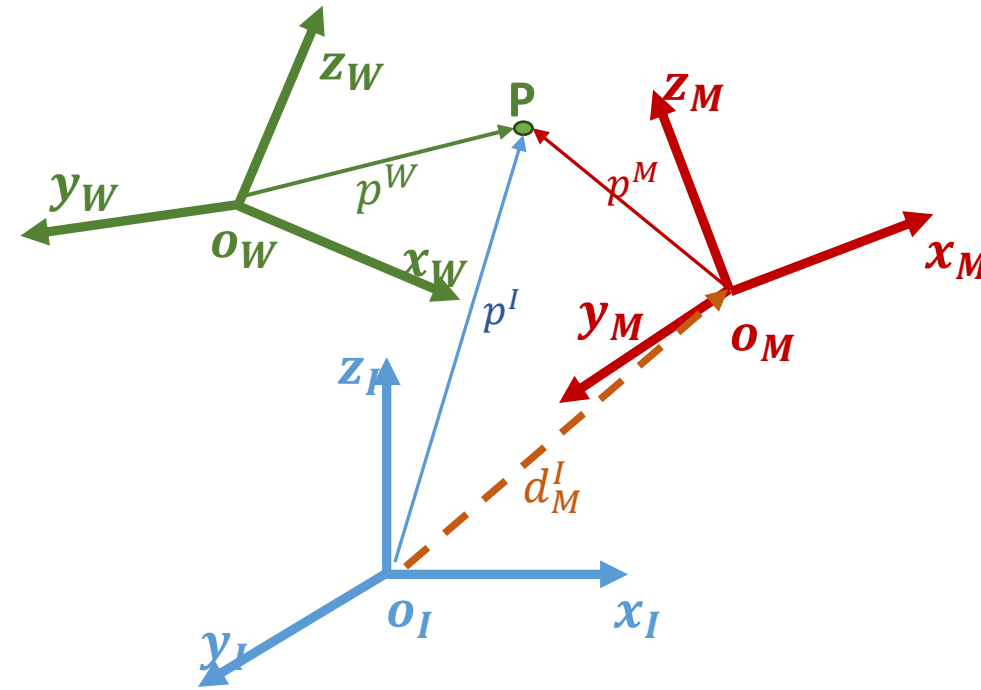
Rigid Body Motion

- For any rigid motion transformation

$$H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}, R \in SO(3)$$

The inverse transformation is

$$H^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}, R \in SO(3)$$



Rigid Body Motion

- To represent rigid motion transformation by a matrix multiplication, we need to augment the vectors p^I and p^M with 1 at the bottom

$$p^{I,H} = \begin{bmatrix} p^I \\ 1 \end{bmatrix} \text{ and } p^{M,H} = \begin{bmatrix} p^M \\ 1 \end{bmatrix}$$

- The vectors $p^{I,H}$ and $p^{M,H}$ are homogeneous representations of p^I and p^M , respectively.
- Thus, the transformation $p^I = R_M^I p^M + d_M^I$ is equivalent to the homogeneous matrix equation $p^{I,H} = H_M^I p^{M,H}$.

